

IBM Cloud Pak for Business Automation Demos and Labs 2026

Configuration and Usage of IBM Content Assistant with Cloud Pak for Business Automation

V 1.0s (for CP4BA 25.0.1 with IBM Content Assistant 1.0.x)

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1 Introduction

1.1 IBM Content Assistant

IBM Content Assistant (ICA) provides natural language insights about business content in IBM FileNet Content Manager repositories. The Content Assistant helps to quickly find information across large volumes of content and saves time by avoiding manual, repetitive review of document content.

IBM Content Assistant includes generative AI capabilities that can summarize document content, compare documents and versions, and provide accurate answers to questions within the context of the selected business content in an IBM FileNet Content Manager repository.

These capabilities are based on the IBM watsonx.ai provided Large Language Models (LLM) that are coupled with context-specific Retrieval-Augmented Generation (RAG). The RAG augments the LLM with context-specific knowledge base and information that is hosted in the IBM FileNet Content Manager repository.

IBM Content Assistant (ICA) provides these capabilities in a secure way, by ensuring that any user running queries will get information only from those documents in the repository, to which this user has access.

1.2 Architectural Overview

The IBM Content Assistant architecture is composed of three primary components:

1. The **ICA plugin** for IBM Content Navigator (ICN) aka IBM Automation Navigator exposes the capabilities of ICA to the end-user by embedding a chat interface into ICN.
2. The **FNCM Gen AI Connector Service** also referred to as **ICA Server** is responsible for transforming documents into vector embeddings, executing semantic search queries against a vector index, and perform the interactions with the underlying Large Language Model (LLM) hosted by IBM watsonx.ai.

Document ingestion involves converting unstructured content into high-dimensional vector representations, enabling efficient similarity search and contextual retrieval. These embeddings are stored in a Vector database, which can run on premises, allowing customers to maintain full control over data locally, or operate as a SaaS-managed service.

3. Two specialized **FileNet Object Store Add-ons** that connect an Object Store in a FileNet Content Platform Engine environment to the ICA Server.

The **5.6.0 Persistent Text Extract Extensions** add-on handles the following aspects:

- Generate text extracts from document content in your repository
- Store the extracted text as Annotations.

The **5.7.0 Gen AI Extensions** add-on handles the following aspects:

- **Expose ICA's capabilities via FileNet Content Engine extensions**

ICA is seamlessly integrated by utilizing standard CPE capabilities like document classes, subscriptions, event handlers and so on.

All capabilities are available through both the Java API and a GraphQL interface, that are also used by the ICA plug-in for Navigator but also support flexible integration into custom applications.

- **Trigger vectorization and AI summary generation and storage**

When a document is to be indexed, the extracted document text is sent for vectorization and AI summary generation to the ICA Server.

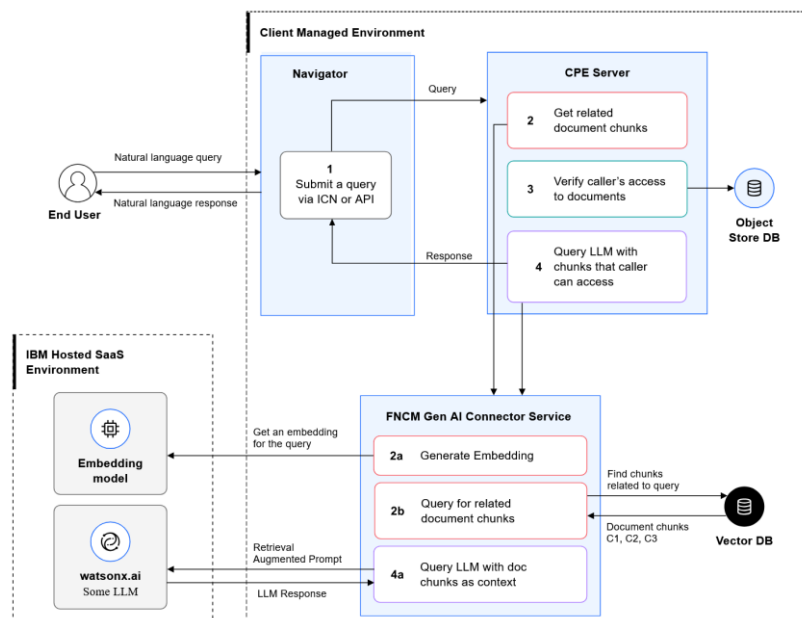
For each document the following is stored:

- **Indexing metadata:** Status and timestamp indicating when the document was ingested into the vector index and an AI summary was generated.

- **AI-generated summary:** A concise abstract generated by the configured LLM hosted by watsonx.ai.
- **Facilitate the execution of ICA queries**
When a user submits a query to the IBM Content Assistant a query object is created in the Content Platform Engine (CPE) object store.

The add-on reacts on query objects being created in the Content Platform Engine (CPE) object store and performs a series of calls to the Gen AI connector service to produce a response:
 - Get document chunks that are related to the query from ICA Server.
 - Submit the user question and document chunks to the LLM hosted by IBM watsonx.ai to generate a response.
- **Ensure that only chunks for document are fed into the LLM which the user can access**
As a part of either determining the documents chunks related to the query or when the whole document is sent to the LLM, it verifies that the user executing the request has at least VIEW_CONTENT access for the specified or identified documents.

Below picture contains the above-mentioned components and a respective call flow when a user performs an ICA query:



For more information, refer to the Overview section in the documentation for IBM Content Assistant <https://www.ibm.com/docs/en/content-assistant?topic=assistant-overview-content>.

1.3 IBM Content Assistant Form-factors

IBM Content Assistant is available as a SaaS ([documentation](#)) offering or can be deployed in parallel of a Cloud Pak for Business Automation or a FileNet Content Platform Engine deployment, called IBM Content Assistant Client Managed Service ([documentation](#)).

The release cycles for both form-factors are not completely aligned. Depending on how ICA is made available in the environment you are using that can result in:

- the chat UI looking slightly differently
- the LLMs and the various prompts for these LLMs being used may differ
- bugs potentially existing in one but not the other form-factor
- Feed back icons (thumbs up/down) not available for the Customer Managed Service.

In addition, some capabilities may be available in one but not the other form-factor altogether.

1.4 Remarks regarding Screenshots and Results

IBM Content Assistant is an innovative approach to make the knowledge that resides in enterprise repositories accessible by using Large Language Models. As such it exists in the very dynamic and rapidly evolving space of generative AI. This means that the ICA itself and the components it relies on (e.g. LLMs) evolve and change quickly.

In relation to this lab that means that:

- Screenshots may look slightly different.
- The sample queries provided may not result in the exact output as seen in the screenshots.
- Executing the same query may produce different output depending on the form-factor, and other factors.

A general word of caution

In general, a Large Language Model (LLM) based capability does not guarantee that the output is always identical when asking the same question multiple times. So be prepared to see output differing from the screenshots. Feel free to play around with the suggested queries to see how the output changes.

The LLM used for this lab is **llama-4-maverick-17b-128e-instruct-fp8**. Using a different LLM or a different version of the same LLM would result in different responses.

1.5 Lab Overview

This lab demonstrates the key use-cases of IBM Content Assistant. The exercises serve as examples. The API exercises are moreover inspired by the needs of automation projects such as the Client Onboarding application showcased in other labs.

The Content Assistant lab can be performed independently of the other Content labs, i.e. independently of "Setting up FileNet Content Platform Engine for Automation Projects on Cloud Pak for Business Automation", "Interfacing FileNet Content Platform Engine with GraphQL on Cloud Pak for Business Automation" and "Introduction to Business Automation Navigator in Cloud Pak for Business Automation".

Exercise "End-user interface" demonstrates the user interface of IBM Content Assistant in IBM Content Navigator aka IBM Business Automation Navigator. It shows how IBM Content Assistant presents itself to the end-users and how it can be used to determine facts from documents in the repository via a chat interface, without the need to open the documents and search for the information. Questions on single documents can be raised directly from the Browse tool of IBM Content Navigator or from the open DaeJa Viewer. Further, questions can also be asked on the whole repository, and it is important that ICA will consider only those documents, to which a given user has access.

Exercise "Administration" provides a deep dive into the administrative aspects of IBM Content Assistant, primarily managed through ACCE (Administrative Console for Content Engine). It focuses on the configuration and behavior of the GenAI Extensions add-on, which enables vector indexing and AI-based summarization for both newly ingested and existing documents via sweeps and event subscriptions.

The exercise demonstrates how to configure:

- **Automatic Text Extraction and Persistent Storage** using a dedicated Object Store add-on to cache extracted text for reuse.
- **Automated Vector Indexing** to register documents in the vector database.
- **Automated Summarization** using an LLM hosted by watsonx.ai to generate document abstracts.

It also shows where these artifacts—such as indexing metadata, summaries, and extracted text—are stored within the FileNet repository, offering insight into the internal workings of IBM Content Assistant.

Exercise "Application Programming Interface" introduces capabilities to create IBM Content Assistant queries through the APIs of FileNet Content Manager. The exercise uses the GraphQL interface for some

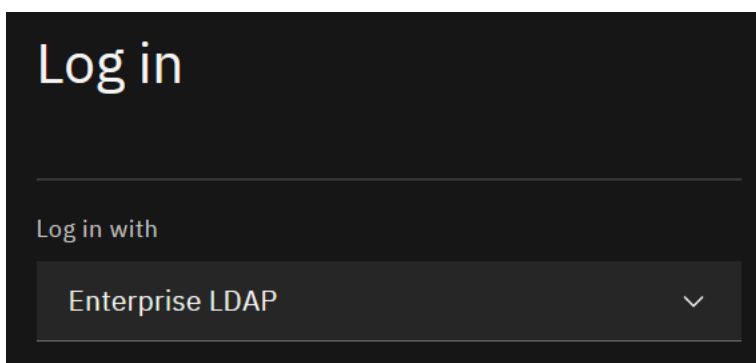
hands-on examples, which can be adapted to the needs of any automation project. It is out of scope for this lab to create and run Java programs. Refer to the [documentation](#) for additional information.

1.6 Lab Setup Instructions

- _1. If you are performing this lab as a part of an **IBM event**, access the document that lists the available systems and URLs along with login instructions. For this lab, you will need to access **IBM Content Services GraphQL**, the **IBM Content Services ACCE**, and the **IBM Content Navigator - ICN desktop**. Further, you will need to have access to a second account named **icademo**. Request the password from the lab instructors.

If you are performing this lab **self-paced** in an environment, which has been enabled for the Client Onboarding labs including the IBM Content Assistant component, you can find the available URLs and information about the icademo user in the **ConfigMap “000-client-onboarding-information”** of the IBM Cloud Pack for Automation project in the OpenShift Web Console.

- _2. The different exercises require different components of the IBM Cloud Pak for Business Automation. Paste the respective URL used into your web browser. The first exercise will use the **IBM Content Navigator - ICA desktop**.
- _3. If you haven’t logged on before, the browser will be redirected to a Log in page. Select **Enterprise LDAP** log in option.



- _4. **Enter** your **Username** (usrXXX) and **Password** as described in step 1 and **click Log in**.

Only when stated explicitly in the lab instructions, use the **icademo** account.

- _5. After successful authentication you will be redirected to the **IBM Content Navigator ICN desktop**. IBM Content Navigator will show the Home page with a tile for each feature configured for the desktop.

2 Exercise: End-user interface

2.1 Introduction

This exercise demonstrates the user interface of IBM Content Assistant (ICA) in IBM Content Navigator (ICN) aka IBM Business Automation Navigator. It shows how IBM Content Assistant presents itself to the end-users and how the chat interface can be used to determine facts from documents in the repository, without the need to open the documents and search for the information. This is achieved through the IBM Content Assistant plugin for IBM Content Navigator which needs to be enabled for a desktop.

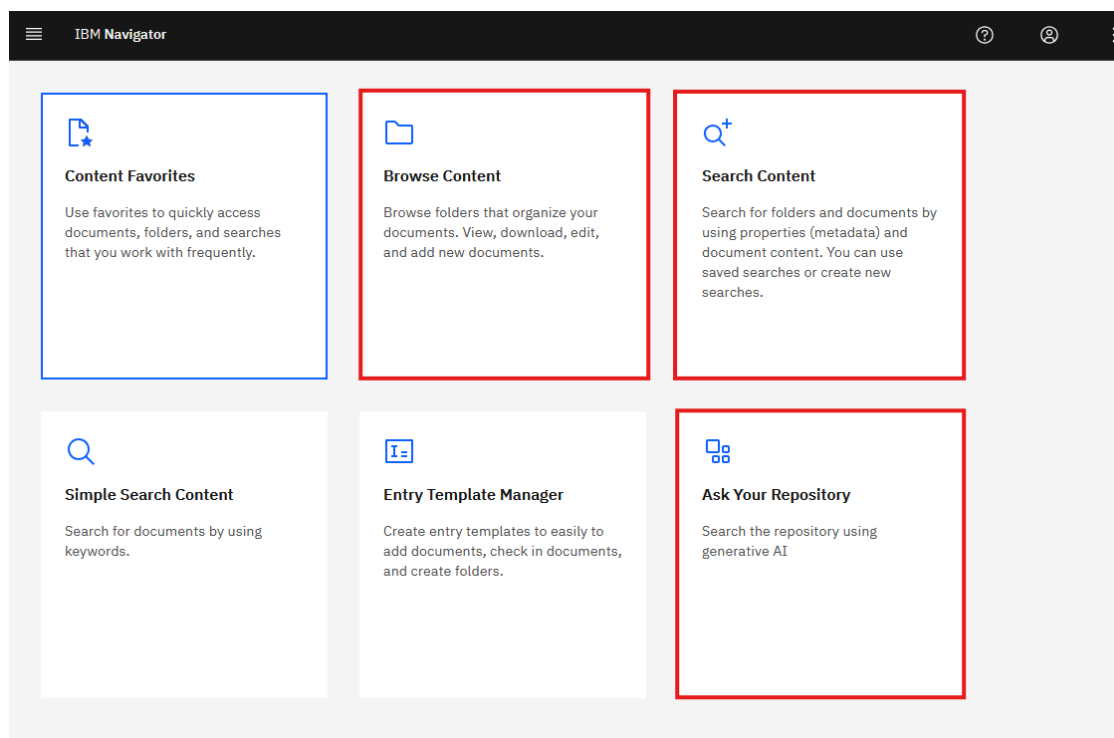
Questions on single documents can be raised directly from the Browse or Search tools of IBM Content Navigator or from the open IBM DaeJa Viewer. IBM Content Assistant is available as a Search Assistant Chatbot in the lower right corner of the browser window. Through the “Ask Your Repository” tool, questions can also be asked on the whole repository. It is an important feature that ICA will consider only the documents, to which a given user has access.

Section 2.2.1 introduces the ICA chat window and its components. Section 2.2.2 focuses on executing queries on single documents and section 2.2.3 on executing queries on multiple documents. Section 2.2.4 demonstrates how to run queries directly from the DaeJa Viewer. IBM Content Navigator is frequently embedded into other applications via bookmark.jsp URLs, which directly opens the DaeJa Viewer for a specific document. In such cases, the IBM Content Assistant chat window remains accessible from within the DaeJa Viewer. Finally, section 2.2.5 shows how to ask questions against the complete repository showcasing that ICA considers only those documents to which a given user has access.

2.2 Exercise Instructions

2.2.1 IBM Content Assistant Chat Window

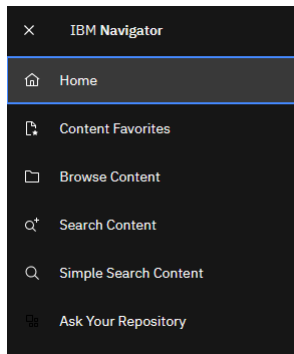
1. After logging in, the IBM Content Navigator Desktop comes up, showing tiles for the Features, which are available on this desktop. For the ICN desktop the following tiles are available.



For IBM Content Assistant foremost the features “Search Content”, “Browse Content” and “Ask Your Repository” are relevant.

The features are also available through the so-called “Hamburger Menu”, this is the icon in the top left corner of IBM Content Navigator. For faster switching, use the Hamburger menu.

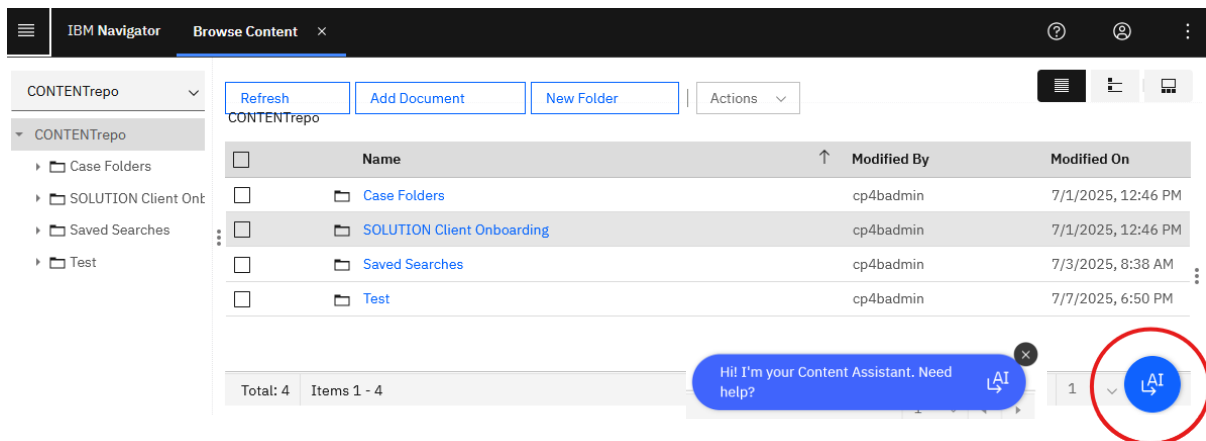
_2. Switch to the **Browse Content** feature.



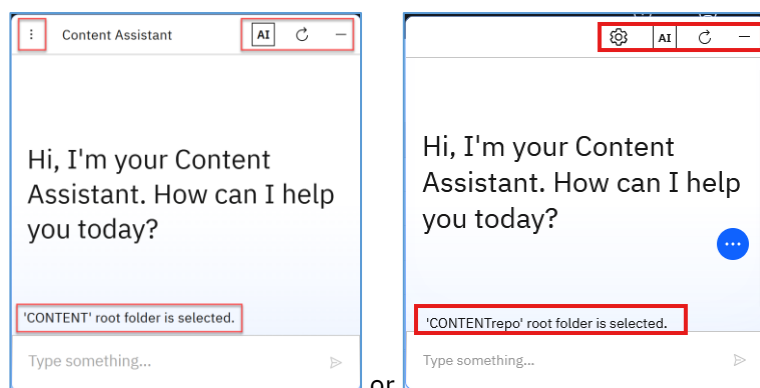
When selected, the “Browse Content” feature shows the root folder of the default FileNet Object Store configured for the desktop, in this case for the CONTENT Object Store. In addition, a dropdown allows you to switch to any other Object Store configured for the desktop.

Depending on the environment you are using the object store name may be shown as CONTENT or CONTENTrepo for the default object store.

When the ICA plugin is enabled, the Browse Content feature shows a round blue button in the lower right corner to invoke IBM Content Assistant. In later versions of the ICA plugin an additional text may be shown in addition, to make that more obvious. Other features of Content Navigator might expose other AI assistants, like the Workplace Assistant when using the Workplace feature.



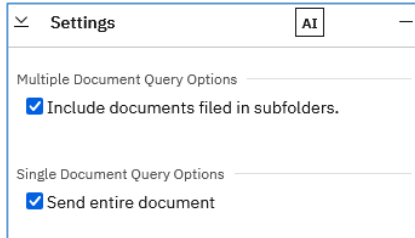
_3. Click on the **AI button** to open the IBM Content Assistant chat window.



Observe the information “CONTENT root folder is selected”. It indicates that as part of a query all documents in the root folder will be passed as context.

Explore the chat window starting with the **toolbar buttons** at the top of the chat window.

- _4. **Click the AI button**, to open a pop-up window displaying information that IBM watsonx is being the underlying AI technology.
- _5. **Open the Settings pane**, that is either accessible via the menu shown when clicking the three vertical dots in the top left corner of the chat window or a wheel icon.



The settings allow you to define whether to **include documents filled in subfolders** in the search prompt or not. This option is enabled by default.

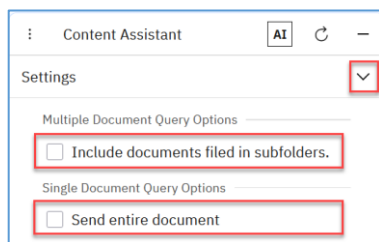
The second option allows you to specify to **Send entire document** when performing single document queries or if only those document chunks are sent to the LLM that are most relevant to the query. This option is enabled by default.

Disabling the option might be useful when the extracted text exceeds 400,000 characters and the entire document cannot be passed as context. In this case, a vector query then automatically retrieves the most relevant document chunks, which are then sent to the LLM along with the prompt to generate a response.

As will be seen later in the lab, the availability of excerpts from the document that was used to generate the response is only available, when a vector query is executed to retrieve the most relevant document chunks, before calling the LLM.

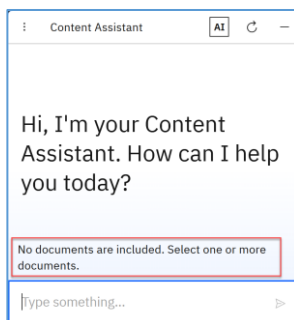
More details are available in the section “Asking questions about a single document” in the documentation <https://www.ibm.com/docs/en/content-assistant?topic=assistant-asking-questions-about-single-document>.

- _6. For now, **disable** both features **Include documents filed in subfolders** and **Send entire document**.

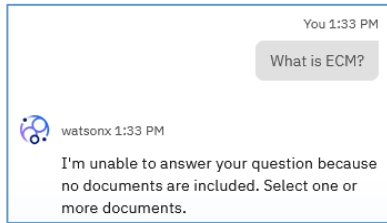


Click on the **symbol** in the left or right of the Settings label to return to the chat window.

Notice that now no documents are included for a query.



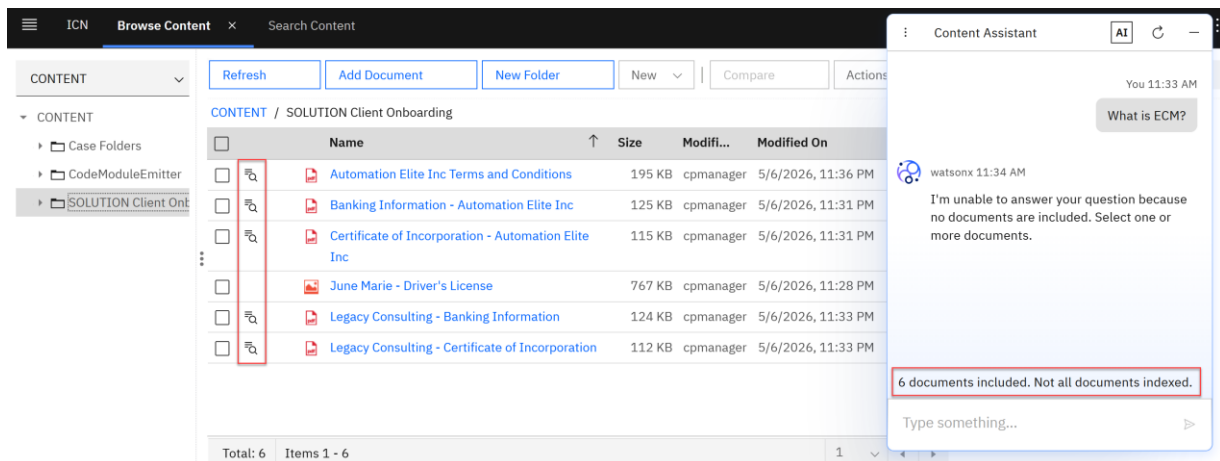
- _7. The **last two buttons restart** the conversation and **minimize** the chat window.
- _8. **Enter in a prompt.** It doesn't matter what you write, you might for example ask **What is ECM?** IBM Content Assistant will refuse to answer, as no documents are included in the query.



Leave the chat window open for the next subsection.

2.2.2 Single Document Queries in the Browse View

- _1. **Select the SOLUTION Client Onboarding** folder in the navigation area on the left side of IBM Content Navigator.



Notice in the Browse view that some documents have an icon on the left side. This icon indicates that the document is vector indexed and can be used in the context of IBM Content Assistant.

Notice further that in the IBM Content Assistant chat window, now the 6 documents contained in that folder are selected for being included in the search query. If no documents are selected in the Browse view, automatically all documents will be included for any search prompt.

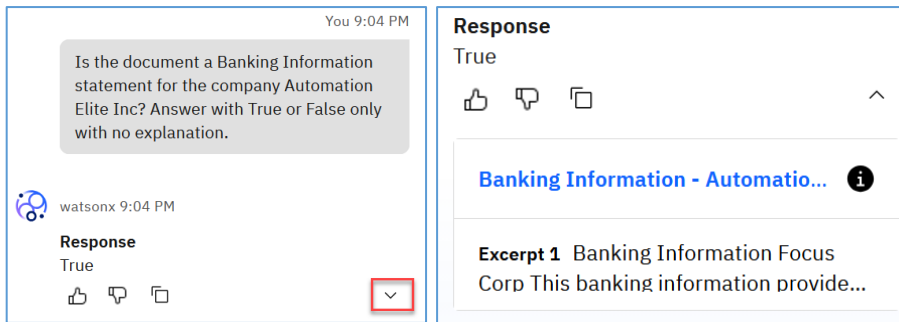
- _2. **Select** the document with the title **Banking Information - Automation Elite Inc.** by checking the checkbox at the beginning of the row. Notice the change in the IBM Content Assistant window.

In the Desktop settings, it can be configured whether document selection checkboxes are shown or hidden. If the checkboxes are hidden, documents can be selected by clicking on them directly. In this case, care should be taken not to click the blue link, as this will open the DaeJa Viewer.

- _3. **Type or copy & paste** the question **Is the document a Banking Information statement for the company Automation Elite Inc? Answer with True or False only with no explanation.**

When post-processing answers from ICA in a custom application, it's a good idea to specify what result would be expected.

You can play around leaving out the **with no explanation** part to see the difference.



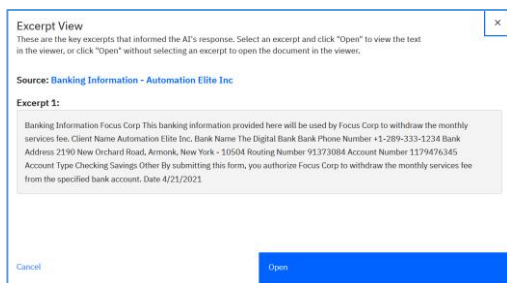
- _4. **Expand** the **twisty** to show the parts of the document (excerpts) from which the answer was derived.

As we have unchecked the option to **Send entire document** in the **Settings**, ICA first performed a vector search to find the most relevant document chunks for our query before sending the query and the document chunks to the LLM.

That is the reason why the twisty is shown.

- _5. **Click** on the **excerpt** to open it in a pop-up window followed by **Cancel**.

You could also open the full document in Daeja Viewer by clicking Open.

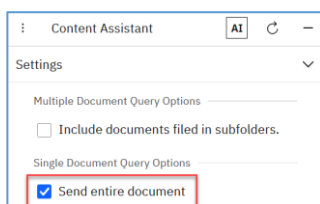


The banking information document is rather short, but in general making use of excerpts helps to validate the response if required, without going through the full document.

- _6. If present, **click** the **thumbs up** icon as the answer is correct.

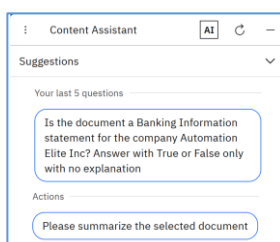
The Thumbs-up / Thumbs-down icons are only available when using ICA SaaS. In ICA Customer Managed Software there is no way to feed back the information to the Large Language Model.

- _7. **Open** the **Settings** again and **enable** the **Send entire document** checkbox before **closing** them again.



- _8. In the ICA chat window, **click** either the **three vertical dots** in the upper left corner and **select Suggestions** or the **blue circle** with the **three dots** to bring up the suggestions pane.

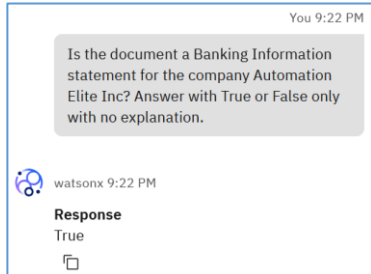
It contains possibly useful further queries or requests, along with a history of your past queries.



_9. **Click** on the **previously executed query** to run it again.

Observe that the response is identical, but the twisty to show the excerpts is not shown.

The reason is that with enabling the **Send entire document** checkbox, ICA no longer first performs a vector search to identify the most relevant document chunks but immediately sends the full document content to the LLM.

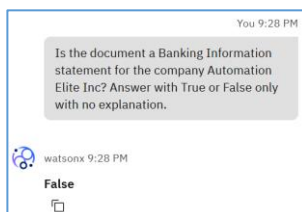


_10. **Select** the **Certificate of Incorporation – Automation Elite Inc** document and **deselect** the **Banking Information – Automation Elite Inc** document.

Refresh	Add Document	New Folder	New	Compare	Actions
CONTENT / SOLUTION Client Onboarding					
☐	Name	Size	Modified By	Modified On	
☐	Automation Elite Inc Terms and Conditions	194 KB	cp4badmin	3/16/2026, 2:33 PM	
☐	Banking Information - Automation Elite Inc	125 KB	cp4badmin	3/13/2026, 12:07 PM	
☒	Certificate of Incorporation - Automation Elite Inc	115 KB	cp4badmin	3/10/2026, 9:14 PM	
☐	June Marie - Driver's License	767 KB	cp4badmin	2/26/2026, 11:32 AM	
☐	Legacy Consulting - Banking Information	124 KB	cp4badmin	3/13/2026, 12:09 PM	
☐	Legacy Consulting - Certificate of Incorporation	112 KB	cp4badmin	3/10/2026, 9:14 PM	

_11. **Re-execute** again the **previous query** by selecting it from the history in the Suggestions, but now on the **Certificate of Incorporation**.

The result will now be **False**. If present, **click** the **thumbs up** again.



_12. **Minimize** the **ICA chat window** using the icon on the top right corner.

_13. **Open/observe** the **properties pane** on the right side of the Content Navigator screen to access the Document properties.

A summary of the document generated by IBM Content Assistant including the date when it was generated is shown.

The screenshot shows the IBM Content Navigator interface. The left sidebar shows the document hierarchy. The main area displays a table of documents. The right sidebar shows the properties of the selected document, "Certificate of Incorporation - Automation Elite Inc". The properties include the document title, the date it was generated (5/13/2026, 9:19 AM), and a summary of the document content.

Name	Size	Modified By	Modified On	Majority
Automation Elite Inc Terms and Conditions	195 KB	cpmanager	5/13/2026, 9:22 AM	2
Banking Information - Automation Elite Inc	125 KB	cpmanager	5/13/2026, 9:19 AM	1
Certificate of Incorporation - Automation Elite Inc	115 KB	cpmanager	5/13/2026, 9:19 AM	1
June Marie - Driver's License	767 KB	cpmanager	5/13/2026, 9:14 AM	1
Legacy Consulting - Banking Information	124 KB	cpmanager	5/13/2026, 9:19 AM	1
Legacy Consulting - Certificate of Incorporation	112 KB	cpmanager	5/13/2026, 9:19 AM	1

Properties

Class: SOLUTION Client Document

Document Title: Certificate of Incorporation - Automation Elite Inc

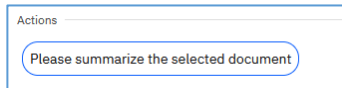
Gen AI Date Indexed: 5/13/2026, 9:19 AM

Watsonx Summary: Automation Elite Inc. is incorporated under the Registration Act of 1967 with corporate identity number 929-3462-4978. The articles of incorporation are effective as of April 27, 2011. The certificate of incorporation was executed on behalf of the Director of Registration Act of 1967 and has been attested. Certificate Number: C-1237890.

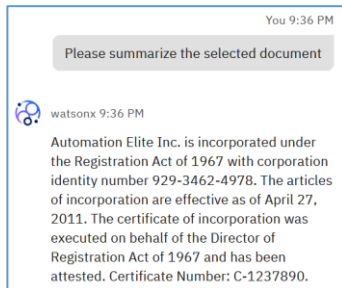
Reference ID: 1000

Client Name: Automation Elite Inc.

- _14. An alternative way to access the information is to open the ICA chat window, open the “Suggestions” and invoke the “Summarize the selected document” action.



Clicking the action returns the summary.



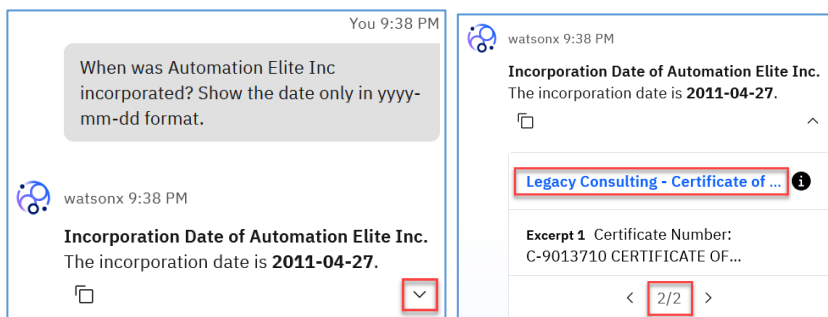
2.2.3 Multi Document Queries in Browse View and Searches

Often ICA will be used to derive information from multiple documents or even many documents. A multi-document query can include up to 1000 documents. Beyond that, a query would better be run against the complete Object Store.

- _1. **Select both Certificates of Incorporation.**

☐	Name	↑	Size	Modifie...	Modified On	Majo...
☐	Automation Elite Inc Terms and Conditions		194 KB	cp4badmin	3/16/2026, 2:33 PM	2
☐	Banking Information - Automation Elite Inc		125 KB	cp4badmin	3/13/2026, 12:07 PM	1
☒	Certificate of Incorporation - Automation Elite Inc		115 KB	cp4badmin	3/10/2026, 9:14 PM	1
☐	June Marie - Driver's License		767 KB	cp4badmin	2/26/2026, 11:32 AM	1
☐	Legacy Consulting - Banking Information		124 KB	cp4badmin	3/13/2026, 12:09 PM	1
☒	Legacy Consulting - Certificate of Incorporation		112 KB	cp4badmin	3/10/2026, 9:14 PM	1

- _2. In the chat window type in or **copy & paste** this question: **When was Automation Elite Inc incorporated? Show the date only in yyyy-mm-dd format.**¹



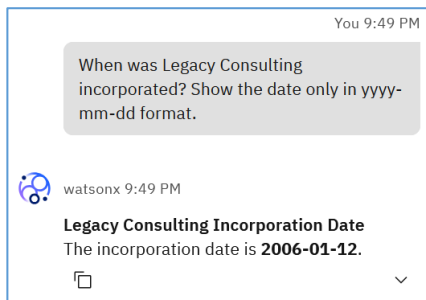
In addition to the correct answer, the twisty is shown providing access to the documents and excerpts that were used to generate the response.

As this is a multi-document search, ICA as a first step ICA performed a vector search to identify the most relevant chunks in the documents selected before passing these to the LLM.

¹ Answers from the IBM Content Assistant, basically any technology based on Large Language Models might vary, influenced by many factors. You can experiment with varying the query, to see how the responses change.

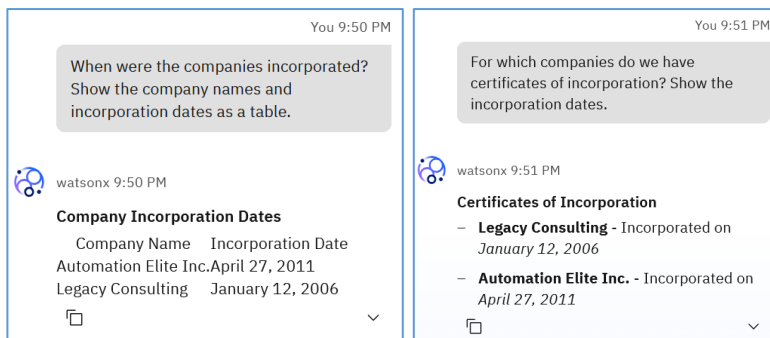
It might seem counterintuitive at first that the Certificate of Incorporation for both Automation Elite Inc. and Legacy Consulting is provided as relevant documents. Yet looking at the query and considering how the vector search works unveils that because of the word “incorporated” in the query, both documents were identified to be relevant as they both contain the words “incorporation” and “incorporated”. Yet in the second stage, the LLM was still able to identify the right document, the one referring to Automation Elite Inc., and to derive the correct incorporation date as well as format it in the requested format.

- _3. **Copy and paste** the same question for Legacy Consulting: **When was Legacy Consulting incorporated? Show the date only in yyyy-mm-dd format.**



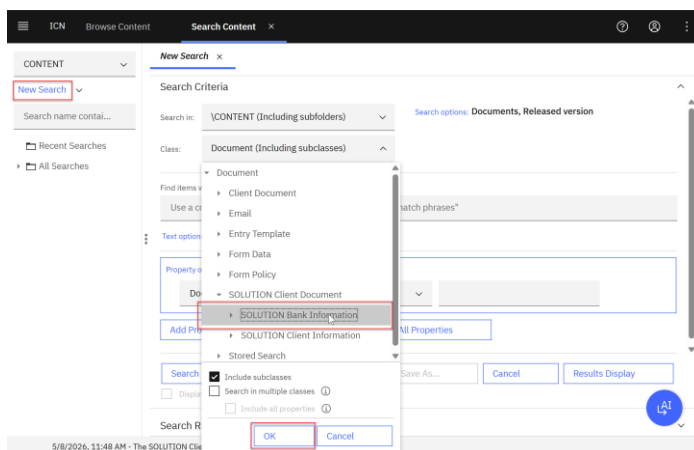
Again, the correct result is provided.

- _4. Explore different queries for finding the incorporation dates for all companies, such as: **When were the companies incorporated? Show the company names and incorporation dates as a table.** or **For which companies do we have certificates of incorporation? Show the incorporation dates.**



IBM Content Assistant can also be used with results of a search.

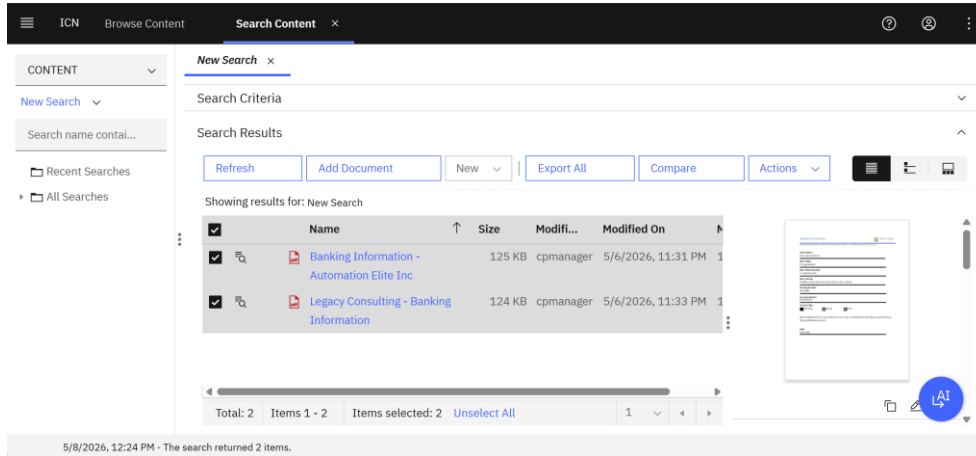
- _5. In Content Navigator use the Hamburger menu in the top left corner to **select** the **Search Content** feature.
- _6. **Click** on **New Search**, then **select** the **SOLUTION Bank Information document class**, as shown below. **Click OK** to close the class selection window.



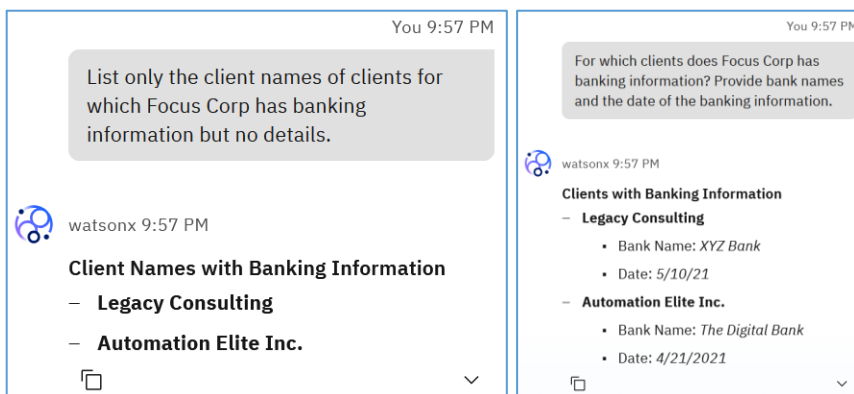
_7. Click the **Search** button to run the search.

_8. In the Search Results **select all documents**.

To do that, **click the checkbox in the table heading row**. However, if you select no documents, it will also default to search all documents, but summarization will only operate on selected documents.

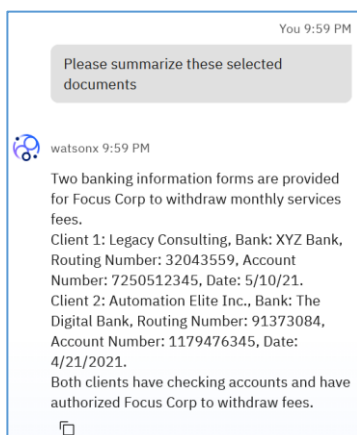


_9. Create a query which will list the company names along with the names of their banks, such as **List only the client names of clients for which Focus Corp has banking information but no details.** or **For which clients does Focus Corp has banking information? Provide bank names and the date of the banking information.**



_10. Open the **suggestions**. Select the query **Please summarize these selected documents**. IBM Content Assistant will provide an ad-hoc summary for the selected documents.

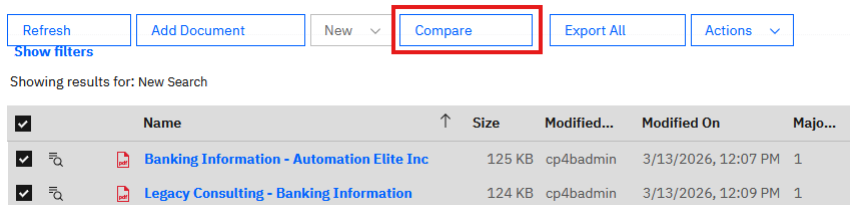
Find more information in the documentation on page: <https://www.ibm.com/docs/en/content-assistant?topic=uca-accessing-watsonx-ad-hoc-summary-your-document-selection>.



IBM Content Assistant can also compare documents. So, let's see the differences between the documents. For this purpose, the IBM Content Assistant ICN add-on exposes a Compare function, which has been added to the ICN desktop.

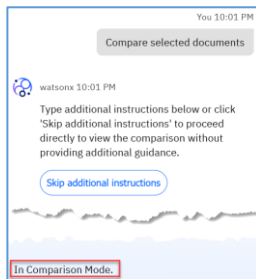
See “Configuring the Document Compare menu action” in the documentation under this URL

<https://www.ibm.com/docs/en/content-assistant?topic=comparison-configuring-document-compare-menu-action>, for further information.



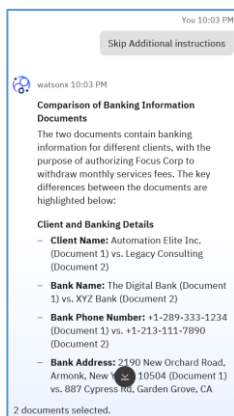
_11. Click the **Compare** button.

The ICA chat window opens, in case it is closed, and gives the opportunity to provide additional instructions for the comparison. In addition, it indicates that comparison mode is enabled.



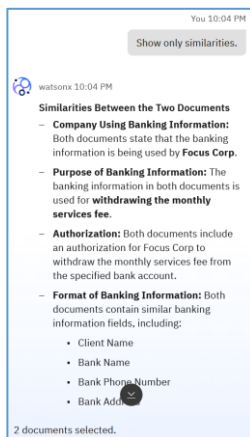
_12. Click on the **Skip additional instructions** button.

Study the resulting output.



In case ICA responds that it cannot respond due to “no relevant document available to provide context”, please retry, or proceed to the next query and then come back to it.

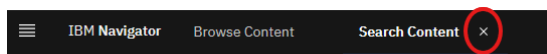
_13. Repeat the **Comparison**, but this time **provide** a comparison instruction **Show only similarities**.



_14. **Deselect one of the documents** and observe that the Compare button will be disabled. If is similarly disabled of more than two documents are selected in the Search or Browse view.

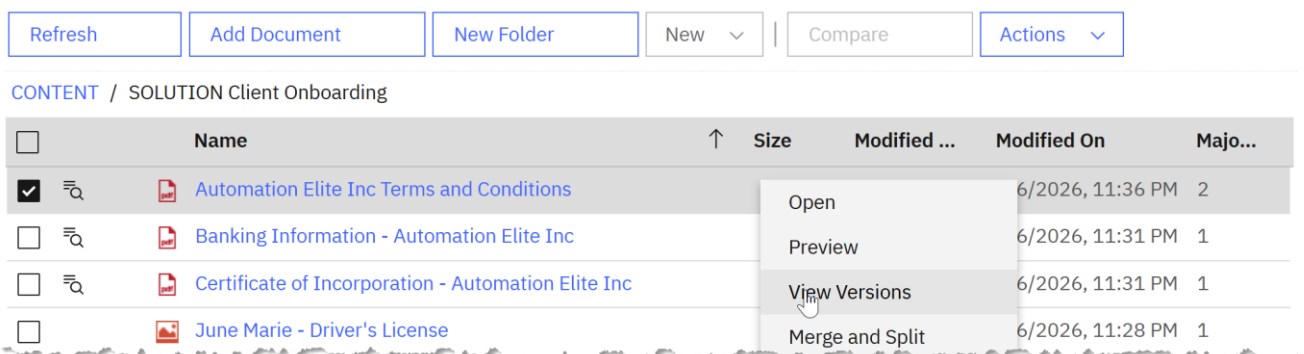


_15. **Click** on the small **x icon** on the right side of the “Search Content” tab in the Navigator title bar, which will close the “Search Content” tab. The previously active “Browse Content” tab is displayed again.



_16. **Right-click** on the **first document** and **click** on the **View Versions** context menu entry.

This is another feature introduced by the IBM Content Assistant add-on. It also needs to be configured to appear in the Content Navigator menus. The feature can be found also in the “Actions” drop down menu. Find a detailed description on the page “Configuring the View Version menu action” in the documentation <https://www.ibm.com/docs/en/content-assistant?topic=comparison-configuring-view-version-menu-action>.



_17. Click on the **View Versions** context menu entry.

The Browse menu changes to show the Version Series of the selected document.

Refresh

New ▾

Compare

Actions ▾

CONTENT / SOLUTION Client Onboarding / Automation Elite Inc Terms and Conditions (Version Series)

☐	Name	Version St...	Major ...	Minor ...	Modified By	Modified On
☐ 	 Automation Elite Inc Terms and Conditions	Released	2	0	cpmanager	5/6/2026, 11:36 PM
☐ 	 Automation Elite Inc Terms and Conditions	Superseded	1	0	cpmanager	5/6/2026, 11:33 PM

Let's find out about the differences between the two versions.

_18. Select both and observe that the **Compare** button gets enabled.

Refresh

New ▾

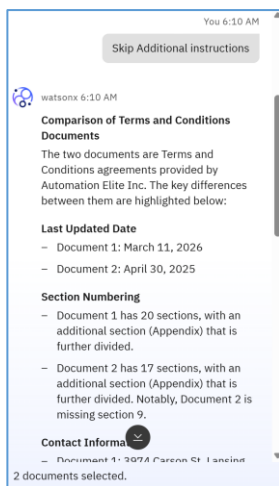
Compare

Actions ▾

CONTENT / SOLUTION Client Onboarding / Automation Elite Inc Terms and Conditions (Version Series)

☒	Name	Version St...	Major ...	Minor ...	Modified By	Modified On
☒ 	 Automation Elite Inc Terms and Conditions	Released	2	0	cpmanager	5/6/2026, 11:36 PM
☒ 	 Automation Elite Inc Terms and Conditions	Superseded	1	0	cpmanager	5/6/2026, 11:33 PM

_19. Click the **Compare** button, give no additional instructions and observe the detailed comparison between the documents.



Assume you were requested to provide a comparison of the document versions to your manager.

Providing a screenshot of the ICA chat window is not an option. Instead, click the copy icon  below the comparison.

Paste the comparison into a new document. You will observe that the text contains html tags used to properly format the text in the chat window.

```
<h4>Comparison of Terms and Conditions Documents</h4>

The two documents are Terms and Conditions agreements provided by Automation Elite Inc. The key differences between them are highlighted below:

<h5>Last Updated Date</h5>
<ul>
  <li>Document 1: March 11, 2026</li>
  <li>Document 2: April 30, 2025</li>
</ul>

<h5>Section Numbering</h5>
<ul>
```

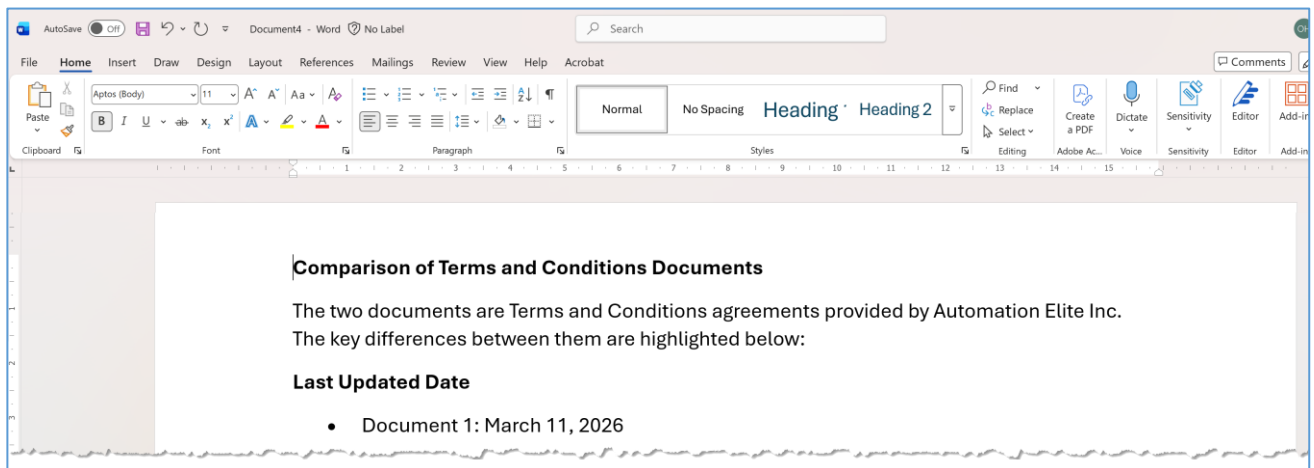
To bring this formatting into a tool like Word, in a text editor, surround the text with `<html>` `<body>` and `</body>` `</html>` tags, save the file as html document.

```
<html>
<body>
<h4>Comparison of Terms and Conditions Documents</h4>

The two documents are Terms and Conditions agreements provided by Automation Elite Inc. The key differences between them are highlighted below:

<h5>Last Updated Date</h5>
<ul>
<li>Document 1: March 11, 2026</li>
<li>Document 2: April 30, 2025</li>
<li>The appendices in Document 1 are more repetitive than those in Document 2.</li>
</ul>
</body>
</html>
```

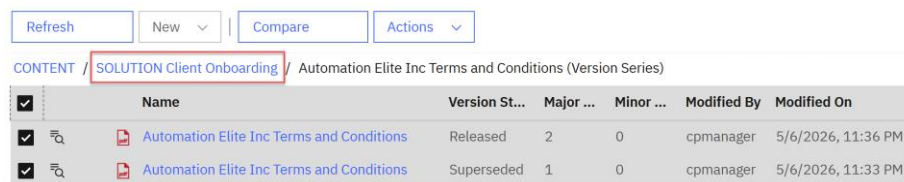
Finally, open it in a web browser, copy the content and then paste it into Word. Word will then be able to reproduce the formatting.



2.2.4 Single Document Queries from the DaeJa Viewer

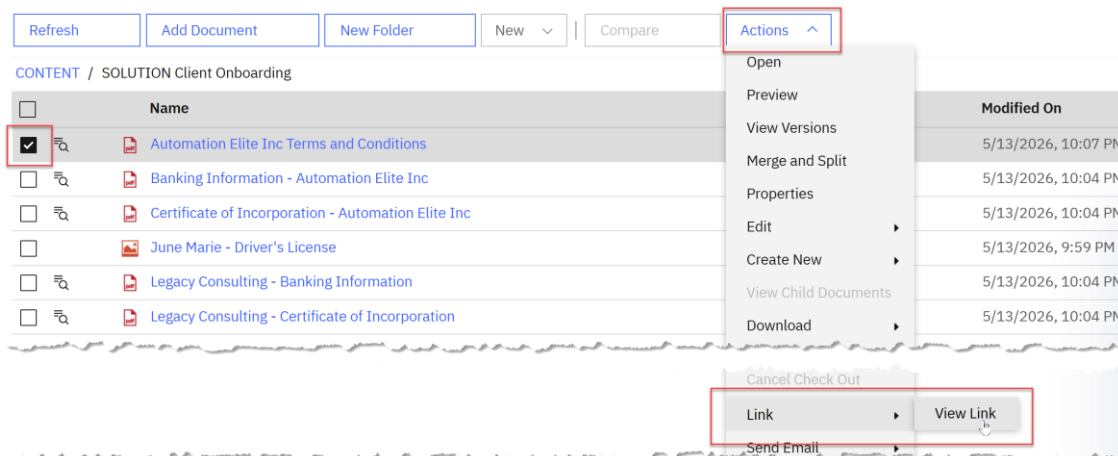
Business solutions, including the Client Onboarding solution, often embed the DaeJa Viewer. Technically this is done by invoking IBM Content Navigator with some additional parameters, which will make it open a specific document with the DaeJa Viewer, instead of opening the complete Content Navigator window. IBM Content Assistant can also be directly used from the DaeJa Viewer window.

- _1. If you did the previous exercise, you can get back to the Browse view with the “SOLUTION Client Onboarding” folder opened by clicking the folder name in the so-called “breadcrumbs”.

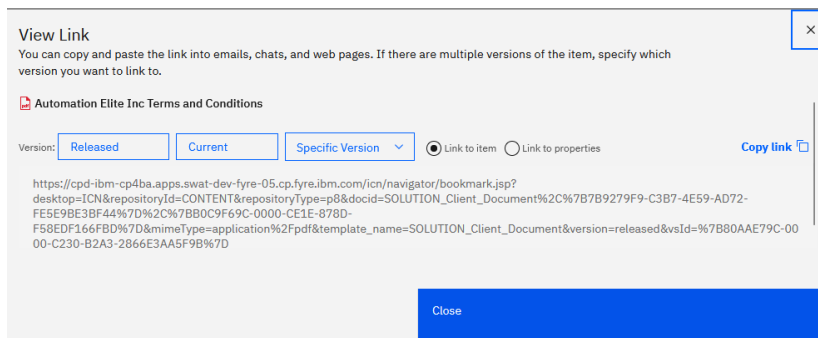


You can also select the Browse view in the Hamburger menu on the top left and navigate to the folder again.

- _2. **Select** only the **Automation Elite Inc Terms and Conditions** document. Then **open** the **Actions** menu and **select View link** from the **Link** menu entry.

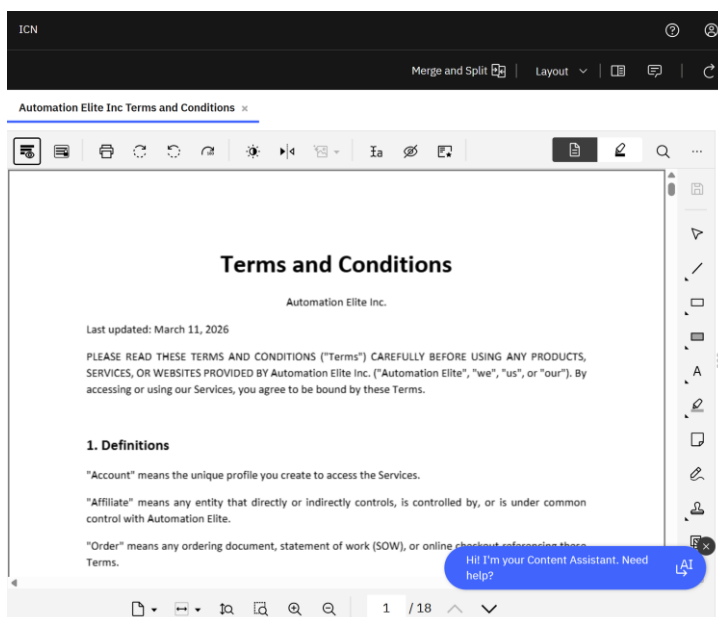


- _3. The “View Link” window opens. Don’t change any options and **click on Copy Link**. Then **close** the **window**.

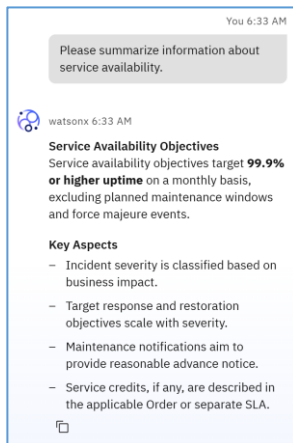


- _4. **Open a new tab** in your browser and paste the link into the address bar. The DaeJa Viewer will open with the selected document, and the ICA chat window button is showing in lower right corner.

Alternatively, you could also just clicked on the document to open it in the DaeJa Viewer.
If you experience any issues with opening the bookmark.jsp URL in a new browser tab, just click the document to open it in DaeJa Viewer.



- _5. Use the chat window to **ask a question** about the document, for example **Please summarize information about service availability.**



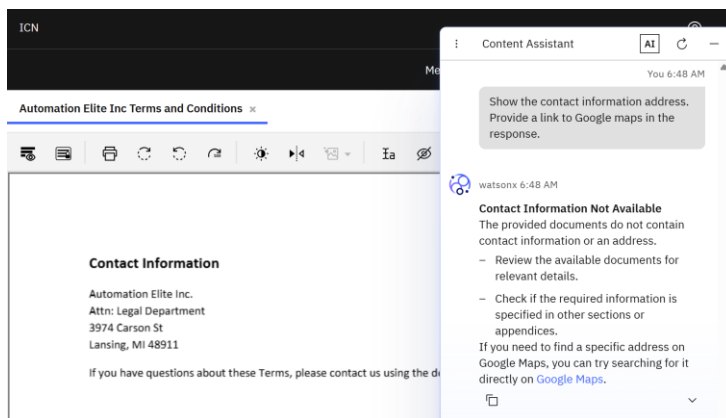
You might be planning a visit to the legal department to discuss details of the document.

- _6. **Scroll down** to the **last page** of the document to find the Contact Information. Let's see where that is.

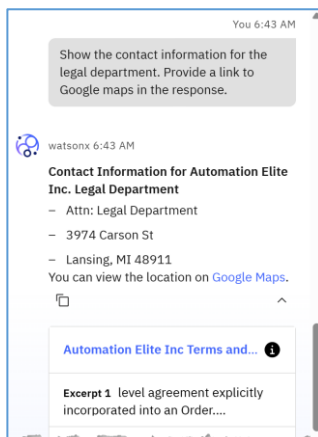
- _7. Temporarily **deselect** the **Send entire document** option in the **Settings**.

- _8. **Enter** the question **Show the contact information address. Provide a link to Google maps in the response.**

In this example the vector search done by ICA has not identified the last page with the actual contact information as a relevant document chunk. Therefore, the LLM could not generate the response containing the address.



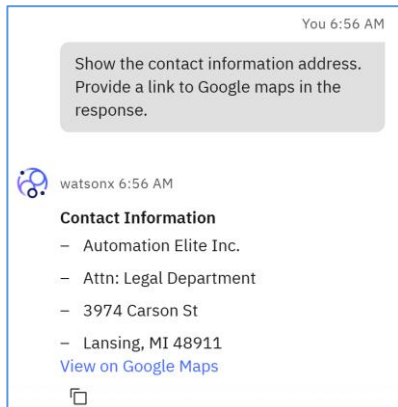
- _9. Enter the slightly more specific question **Show the contact information for the legal department. Provide a link to Google maps in the response.**



With additional information about the legal department, the vector search identified the document chunk that contains the address and passes that to the LLM which in turn creates the correct response.

Optionally expand the excerpts view and check which excerpt contains the address of the legal department.

- _10. **Right click** on the **link** and **open it** in a **new tab**. Google Maps opens at the specified address. Close the tab again.
- _11. **Change** back the **Send entire document** option in the **Settings** to **selected**.
- _12. **Re-execute** the question **Show the contact information address. Provide a link to Google maps in the response.**



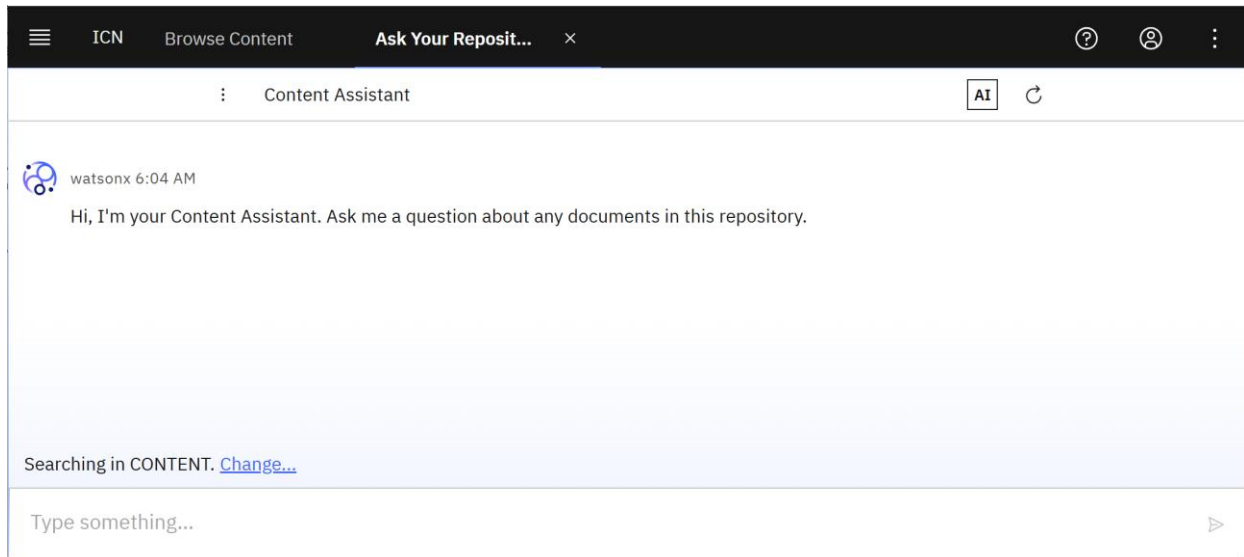
This time, despite being more unspecific, the LLM is able to provide the correct response. The reason is that it has the full document content available for processing.

This shows that in many cases it is beneficial to send the full document content to the LLM without doing a vector search first. This is the reason why ICA has changed the default value for the “Send entire document” option to be selected compared to earlier versions. On the other hand, this results in no document excerpts being available. Depending on the business requirements the user may decide to enable or disable the “Send entire document” option.

2.2.5 Running Queries against an Object Store

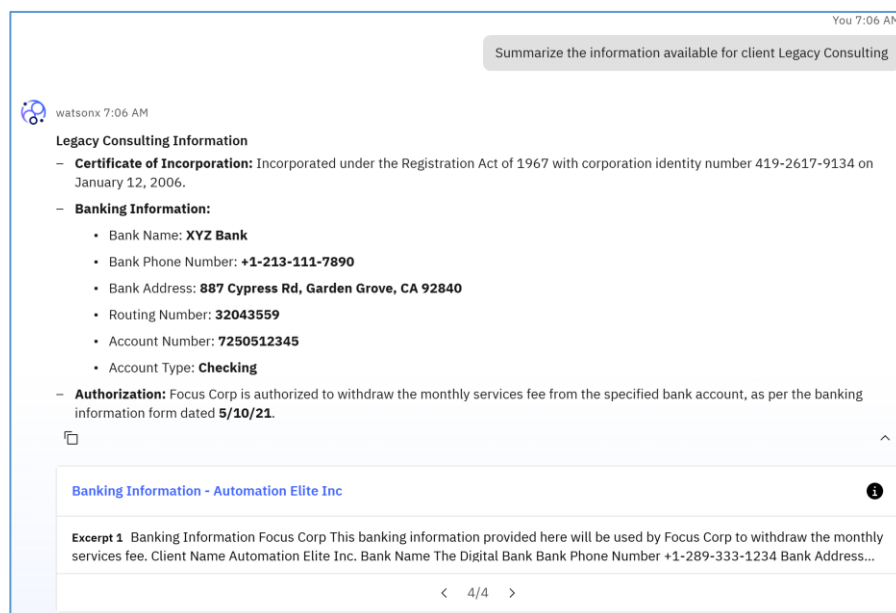
- _1. Use the Hamburger menu in the top left corner to **select** the feature **Ask Your Repository**.

An enlarged ICA chat window will be shown. Implicitly, all documents available to the logged-on user are included when asking a question.



Above the search prompt field, the repository being searched in is listed. If more than one repository enabled for ICA is available in the desktop, a “Change...” link is displayed. By clicking on that the repository can be changed.

- _2. **Enter** this query **Summarize the information available for client Legacy Consulting** as an example.



Observe that the answer contains information both from the Banking Information and the Certificate of Incorporation for Legacy Consulting.

- _3. **Expand** the **documents** section.

Notice that more than just the two documents for Legacy Consulting were included to generate the response, but also two documents that are for Automation Elite Inc.

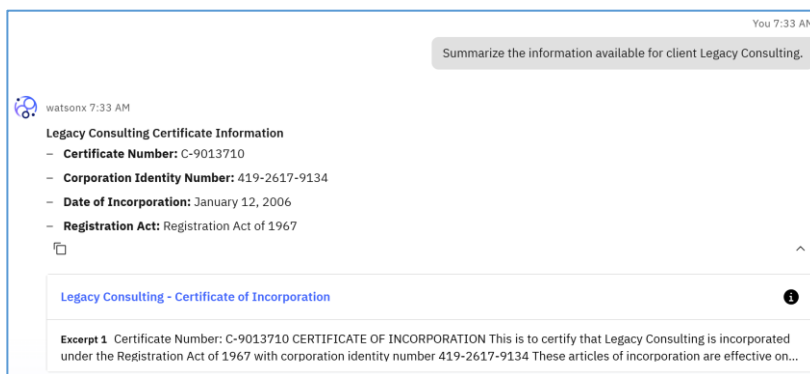
ICA first performed a vector search based on the query to identify those documents and document chunks from the whole repository that are most relevant to the question before passing these on together with the question to the LLM. The mention of words like “client” that also appear in the other two documents resulted in them to be included in the documents passed to the LLM. Nevertheless, the LLM was then able to create the correct response by only considering those documents that are really related to Legacy Consulting.

- _4. In **another browser, or in a private window** of the same browser, **login** to the **IBM Content Navigator ICA** desktop again.

In this session use the **icademo** user using the password provided as part of the access information for the environment.

Importantly, the **icademo** user has **only access** to the **Certificate of Incorporation for Legacy Consulting** but **no access** to other documents including the **Banking Information Statement for Legacy Consulting**.

- _5. In that window, **open** the **Ask Your Repository** feature.
- _6. **Ask** the **same question** as above **Summarize the information available for client Legacy Consulting**.
- _7. **Observe** the answer to the icademo user is different and **does not contain any banking information** as that document is not available for the icademo user.



- _8. **Check** the documents used to generate the response.

This confirms that only the Certificate of Incorporation for Legacy Consulting was used as this is the only document visible to the icademo user.

One core feature that sets ICA apart from other RAG solutions is, that it ensures that only those documents are used to generate the response that are visible to the logged in user.

3 Exercise: Administration

3.1 Introduction

Communication between the IBM Content Assistant chat interface—whether embedded in IBM Content Navigator or used from a custom application—and the IBM Content Assistant server is handled through the IBM FileNet Content Platform Engine (CPE).

Administration and configuration of ICA are implemented at the Object Store level by deploying an Object Store plug-in into each Object Store that is intended to be integrated with IBM Content Assistant. Each of these Object Stores must be explicitly configured for this integration.

For this architecture to function, the current version of IBM Content Assistant SaaS requires that outbound communication from the FileNet environment to the IBM Content Assistant server components is possible and secured using TLS 1.2 or higher.

When IBM Content Assistant Server is operated as an IBM SaaS offering, communication relies on publicly trusted certificates. In contrast, for custom deployments using IBM Content Assistant – Client Managed Software (ICACMS), self-signed certificates may be used. In such cases, these certificates must be added to the Cloud Pak for Business Automation environment to establish trusted communication between the FileNet platform and the IBM Content Assistant server components.

Within an Object Store, several administrative steps are required to enable IBM Content Assistant features for documents. These steps allow the document text to be extracted and persistently stored, indexed in the IBM Content Assistant vector index database, and—optionally—to generate and store a summary in a document property.

These configurations are applied at the document class level and control how documents are prepared for use by IBM Content Assistant. Not all features must be enabled for every document class. For example, document for a document class may be included in the vector index for question answering without requiring that a summary is generated and stored.

This flexibility allows administrators to tailor the setup based on use cases, storage considerations, and performance requirements, enabling only those capabilities that are needed for a given set of documents.

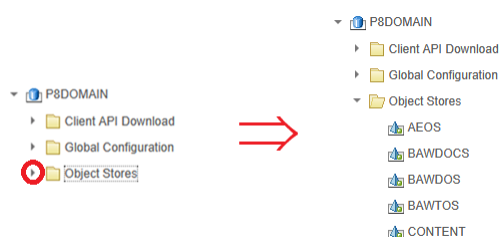
In this exercise, the most important FileNet Components of the IBM Content Assistant will be demonstrated.

3.2 Exercise Instructions

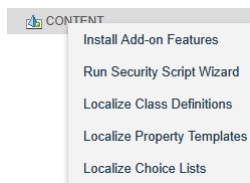
For this exercise, open the Administration Console for Content Engine (ACCE) via the link shared in the environment information. Login with your **usrxxx** and password (replacing xxx by the number of your user).

3.2.1 Content Assistant Add-on and Configuration

- _1. After logging in to ACCE, the Domain Property window is shown in the central area, and a navigation tree on the left side.
- _2. **Click** on the **small triangle** in front of the **Object Stores** label to open the list of Object Stores (which depending on your environment may have additional entries).



_3. **Right-click** on the **CONTENT** Object Store and **select Install Add-on Features**.



_4. A window opens which shows the installed and not-yet installed add-ons for the CONTENT Object Store. Review the list of installed add-ons and find the two add-ons (5.6.0 Persistent Text Extract Extensions & 5.7.0 Gen AI Extensions) that are provided in the context of the IBM Content Assistant.

In FileNet Content Platform Engine 5.7.0, which is used with Cloud Pak for Business Automation 25.0.1, these two add-ons are already pre-installed. For versions 5.6.0 of FileNet Content Platform Engine, the add-ons need to be installed before they can be added to an Object Store. The documentation provides details how this is done, see <https://www.ibm.com/docs/en/content-assistant?topic=engine-installing-content-assistant-add-object-store>.

Installed add-on features

Display Name	Type	Prerequisites
5.2.1 Workplace XT Extensions	Optional	5.2.1 Workplace Base Extensions
5.6.0 Persistent Text Extract Extensions	Optional	
5.7.0 Gen AI Extensions	Optional	
IBM Content Navigator 2.0.3 Entry Template Extensions	Optional	5.2.1 Workplace Templates Extensions 5.2.1 Workplace XT Extensions
IBM Content Navigator 3.0.0 Redaction Extensions	Optional	
IBM Content Navigator 3.0.1 Edit Service Extensions	Optional	

After adding the add-ons to each Object Store which is to be used with IBM Content Assistant, IBM Content Assistant is not yet functional. To enable IBM Content Assistant for an Object Store, it requires a connection to the backend part of IBM Content Assistant and an access code when using ICA SaaS.

_5. **Click** on the **CONTENT Object Store** to open it.

_6. **Click** on **Search** in the left-hand tree, to open the “Saved Searches” window on the right side.

_7. **Click** on the **New Object Store Search** button in the header row of the “Saved Searches” window. The “New Object Store Search” window opens.

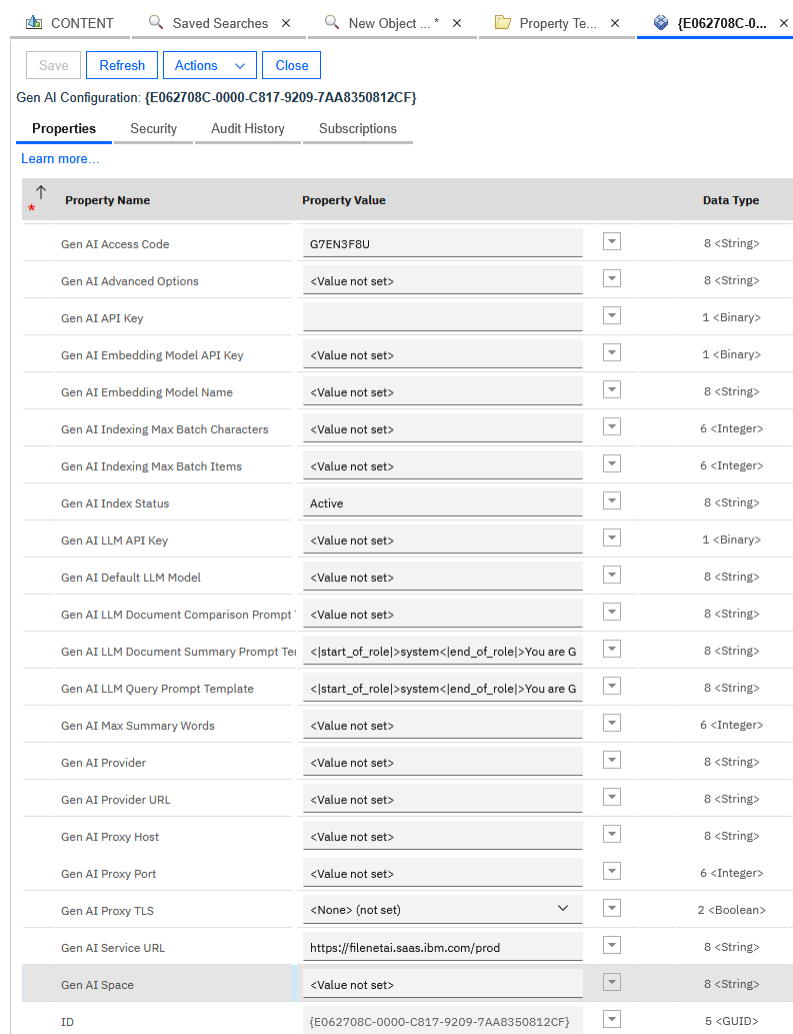
Property	Condition	Value
A	<none>	
B	<none>	
C	<none>	
D	<none>	

_8. For the **Class**, type in or select from the dropdown list the class **Gen AI Configuration**. No other changes are required.

_9. **Run** the search and **dismiss the warning**.

_10. **Click** on the **blue ID value** for the single item found to open the IBM Content Assistant Configuration object.

The properties where the name starts with “Gen AI” store the configuration of IBM Content Assistant.



Property Name	Property Value	Data Type
Gen AI Access Code	G7EN3F8U	8 <String>
Gen AI Advanced Options	<Value not set>	8 <String>
Gen AI API Key		1 <Binary>
Gen AI Embedding Model API Key	<Value not set>	1 <Binary>
Gen AI Embedding Model Name	<Value not set>	8 <String>
Gen AI Indexing Max Batch Characters	<Value not set>	6 <Integer>
Gen AI Indexing Max Batch Items	<Value not set>	6 <Integer>
Gen AI Index Status	Active	8 <String>
Gen AI LLM API Key	<Value not set>	1 <Binary>
Gen AI Default LLM Model	<Value not set>	8 <String>
Gen AI LLM Document Comparison Prompt	<Value not set>	8 <String>
Gen AI LLM Document Summary Prompt Template	<[start_of_role]>system<[end_of_role]>You are G	8 <String>
Gen AI LLM Query Prompt Template	<[start_of_role]>system<[end_of_role]>You are G	8 <String>
Gen AI Max Summary Words	<Value not set>	6 <Integer>
Gen AI Provider	<Value not set>	8 <String>
Gen AI Provider URL	<Value not set>	8 <String>
Gen AI Proxy Host	<Value not set>	8 <String>
Gen AI Proxy Port	<Value not set>	6 <Integer>
Gen AI Proxy TLS	<None> (not set)	2 <Boolean>
Gen AI Service URL	https://filenetai.saas.ibm.com/prod	8 <String>
Gen AI Space	<Value not set>	8 <String>
ID	{E062708C-0000-C817-9209-7AA8350812CF}	5 <GUID>

- _11. The property “Gen AI Index Status” reflects if the Object Store is enabled for communication with the IBM Content Assistant server. The value initially is “Inactive”. When the Object Store is successfully enabled, the value changes to “Active”.
- _12. The property “Gen AI Access Code” is an access code, which grants one FileNet environment access to the IBM Content Assistant Server in the SaaS environment. All Object Stores in one environment can use the same value. Object Stores in different environments will need a different access code. For a Client Managed Software it can have an arbitrary value.
- _13. The property “Gen AI Service URL” needs to be set to the correct endpoint, either of the SaaS instance or the local connection within the OpenShift cluster CP4BA is running for the Customer Managed Software option. It is the URL of the IBM Content Assistant Server to be used for the environment.
In case a Customer Managed Software (ICACMS) is used, the service URL might be using self-signed certificates, which will need to be added to the Cloud Pak for Business Automation specification.
- _14. Locate the “Gen AI LLM Query Prompt Template” property. Click the small arrow downwards beside the value and invoke “Display or Edit Value”. A window opens with the IBM Content Assistant Query Master Prompt, into which any user prompt is embedded when a query is executed. You can copy the value to a text window to more easily read it. Close the window again.

The master prompt can be modified to for example instruct the LLM to provide the output in a specific language, regardless of the language of the query.

For document comparison and summary generation there are other master prompts which can be modified, if needed.

- _15. Other important properties and their use are documented in the IBM Content Assistant documentation. See <https://www.ibm.com/docs/en/content-assistant?topic=assistant-configuring-content-service-parameters-object-store> for details.

They can be used for changing the Large Language and Embedding Models, which are used by IBM Content Assistant, suspending vector indexing, changing the relevancy score for documents, or configuring the most often used default queries.

As pointed out at the beginning of the lab, your environment is configured to use **meta-llama/llama-4-maverick-17b-128e-instruct-fp8** as the Gen AI Default LLM Model.

Gen AI Default LLM Model

meta-llama/llama-4-maverick-17b-128e-instruct

- _16. Don't make any changes to the configuration and close the IBM Content Assistant Configuration Properties window.

3.2.2 Configuring Persistent Storage of Text Extraction

Text Extraction from Documents was provided by FileNet Content Manager already before IBM Content Assistant was introduced. It was used in the process of CBR indexing documents. In FileNet Content Manager 5.6.0 the feature to persistently store the extracted text was introduced. The reason was to save time, primarily when documents are to be CBR indexed AND vector indexed. Consequently, the persistent text extraction is enabled through its own Object Store add-on, as could be seen in the previous subsection.

- _1. Starting with CPE 5.7.0, persistent storage of Text Extracts is configured on the document class level, on the General tab. **Navigate** in the **CONTENT** Object Store to **Data Design -> Classes** and **click** on the **Document** document class to open its properties.

The installation of the Object Store add-on “Persistent Text Extract Extensions” has added the additional property “Persistent Text Extracts enabled” on the General tab.

As you see, it's not enabled for the “Document” Document class.

Object Store: CONTENT

The screenshot shows the 'Document' class definition in the 'CONTENT' Object Store. The 'General' tab is active, displaying various properties for the 'Document' class. The 'Persistent Text Extracts enabled' checkbox is highlighted with a red box, indicating it is currently unchecked. Other visible properties include 'Display name', 'Symbolic name', 'Description', 'Table definition', 'Superclass definition', 'Immediate subclasses', 'Default storage area', 'Default storage policy', 'Default document lifecycle policy', and 'Default replication group'.

Property	Value
Display name	Document
Symbolic name	Document
Description	A single version of a document stored in an object store.
Table definition	
Superclass definition	
Immediate subclasses	Client Document, Code Module, Edit Service Template, Email, Entry Template
Support versioning	☒
Compound document	☐
CBR enabled	☒
Persistent Text Extracts enabled	☐
Default storage area	<None>
Default storage policy	Default Database Storage Policy
Default document lifecycle policy	<None>
Default replication group	<None>

- _2. **Click** on the small **triangle** in front of the **Document** document class in the navigation pane. This shows the subclasses of the “Document” document class.
- _3. **Open** the **SOLUTION Client Document** document class and display its “General” tab. As you see, “Persistent Text Extracts” is enabled for this document class.

The screenshot shows the P8DOMAIN CONTENT interface. On the left, the navigation pane displays a tree structure under 'CONTENT'. The 'Document' class is expanded, showing its subclasses: 'Client Document', 'Code Module', 'Edit Service Template', 'Email', 'Entry Template', 'Preferences Document', 'SOLUTION Client Document', 'Stored Search', 'Workflow Definition', and 'XML Property Mapping Script'. The 'SOLUTION Client Document' class is selected. On the right, the 'Class Definition: SOLUTION Client Document' tab is active, showing the 'General' tab. The 'Display name' is 'SOLUTION Client Document', the 'Symbolic name' is 'SOLUTION_Client_Document', and the 'Description' is 'SOLUTION Client Document'. The 'Table definition' is 'Document'. The 'Superclass definition' is 'Document'. The 'Immediate subclasses' are 'SOLUTION Bank Information' and 'SOLUTION Client Information'. The 'Default storage area' is '<None>', the 'Default storage policy' is 'Default Database Storage Policy', the 'Default document lifecycle policy' is '<None>', and the 'Default replication group' is '<None>'. The 'Support versioning' checkbox is checked, the 'Compound document' checkbox is unchecked, the 'CBR enabled' checkbox is checked, and the 'Persistent Text Extracts enabled' checkbox is checked.

- _4. **Click** the small **triangle** in front of the **SOLUTION Client Document** class to reveal its subclasses.
- _5. **Select** one of its subclasses, e.g. the “SOLUTION Bank Information” class. There you see that the property is enabled but greyed out. This is indicating that it is enabled on a parent document class and cannot be disabled at this level in the class hierarchy.



- _6. For reviewing, where the extracted text is persistently stored, **navigate** to the **Browse -> Root Folder -> SOLUTION Client Onboarding**. **Click** on the **SOLUTION Client Onboarding** folder to display its contents on the right side.

The screenshot shows the P8DOMAIN CONTENT interface. On the left, the navigation pane displays a tree structure under 'CONTENT'. The 'SOLUTION Client Onboarding' folder is selected. On the right, the 'Folder: SOLUTION Client Onboarding' tab is active, showing the 'Contents' tab. The table displays the following data:

	Containment Name	Document Name	Date Created	Created By	Class	Major Version Number
☐	Automation Elite Inc Terms and Conditions.pdf	Automation Elite Inc Terms and Conditions	March 16, 2026 at 2:15:40 PM Central European Standard Time	cp4badmin	SOLUTION Client Document	2
☐	Banking Information - Automation Elite Inc.pdf	Banking Information - Automation Elite Inc	February 26, 2026 at 11:32:35 AM Central European Standard Time	cp4badmin	SOLUTION Bank Information	1
☐	Certificate of Incorporation - Automation Elite Inc.pdf	Certificate of Incorporation - Automation Elite Inc	February 26, 2026 at 11:32:37 AM Central European Standard Time	cp4badmin	SOLUTION Client Document	1
☐	June Marie - Driver's License.png	June Marie - Driver's License	February 26, 2026 at 11:32:39 AM Central European Standard Time	cp4badmin	SOLUTION Client Information	1
☐	Legacy Consulting - Banking Information.pdf	Legacy Consulting - Banking Information	February 26, 2026 at 11:32:41 AM Central European Standard Time	cp4badmin	SOLUTION Bank Information	1
☐	Legacy Consulting - Certificate of Incorporation.pdf	Legacy Consulting - Certificate of Incorporation	February 26, 2026 at 11:32:43 AM Central European Standard Time	cp4badmin	SOLUTION Client Document	1

_7. **Click** on the **second document**, as it uses the class “SOLUTION Bank Information”, a subclass of “SOLUTION Client Document” and has persistent text extraction enabled.

_8. **Click** on the **Annotations** tab.

The screenshot shows the CONTENT application interface. At the top, there are tabs for 'CONTENT', 'SOLUTION CL...', 'SOLUTION Ba...', 'SOLUTION CL...', and 'Banking Inf... (v1.0)'. Below the tabs, there are buttons for 'Save', 'Refresh', 'Actions', and 'Close'. The main header indicates the document is 'Banking Information - Automation Elite I...', Version: 1.0, Status: Released. The 'Annotations' tab is selected, showing a table with columns: ID, Description, and Annotated Content Element (ESN). The table contains one row with ID '{50D5D097-0000-C092-82FD-CBCA97AB4A01}', Description 'Text extract for element 0', and ESN '0'. There are also 'Add' and 'Remove' buttons above the table.

_9. FileNet Content Manager has created an annotation object with the extracted text for this document. To display its content, **click** on the **blue link** to open the Annotation object.

On the “General” tab any error codes are displayed which occurred during text extraction. In this case it succeeded, and there is no value for that property.

The screenshot shows the 'Annotation: {50D5D097-0000-C092-82FD-CBCA97AB4A01}' page. The 'General' tab is selected, showing fields for 'Annotated Object' (Banking Information - Automation Elite Inc), 'Description' (Text extract for element 0), and 'Indexing failure codes' (empty). There are buttons for 'Save', 'Refresh', 'Actions', and 'Close' at the top.

_10. **Click** on the **Content Elements** tab of the Annotation object. **Select** the single **content element**.

The screenshot shows the 'Content Elements' tab of the Annotation object. A 'Content Element' dropdown menu is open, showing a single element selected. The table below has columns: ESN, Class, Content Type, Retrieval Name, Content Location, Size, and Thumbnail. The selected element has ESN '0', Class 'ContentTransfer', Content Type 'text/plain', Retrieval Name 'extract0.txt', Content Location, Size '574.00 bytes', and a Thumbnail.

_11. **Click** on **View/Download** in the **Content Element** menu to download the text file with the extracted text. It is downloaded by the browser, and you can open it to display its content. **Close** it again.

_12. Back in ACCE, **switch back** to the **content** of the folder **SOLUTION Client Onboarding**. Find the Annotation object for “June Marie - Driver's License” and open it. Check the “Content Elements” tab.

The screenshot shows the 'Annotation: {50D5D097-0000-C46A-A5D4-CA6B1D38C887}' page. The 'Content Elements' tab is selected, showing a 'Content Element' dropdown menu. The table below has columns: ESN, Class, Content Type, Retrieval Name, Content Location, Size, and Thumbnail. The message 'No items to display.' is shown below the table.

FileNet Content Manager has created an Annotation object without content to indicate that the text extraction has not resulted in any text. By storing the Annotation, text extraction for such documents will not be tried repeatedly, for documents without usable text in it. Close the Annotation object again.

_13. In the navigation area, **navigate** to **Sweep Management** and **expand Queue Sweeps**.

- ▼ Sweep Management
 - ▶ Background Search Sweeps
 - ▶ Job Sweeps
 - ▶ Policy-Controlled Sweeps
 - ▼ Queue Sweeps
 - Abandoned Content Deletion Sweep
 - Content Deletion Sweep
 - Content Replication Sweep
 - Gen AI LLM Operations Queue Sweep
 - Gen AI Vector Indexing Queue Sweep
 - Text Extraction Queue Sweep
 - Thumbnail Request Sweep
 - ▶ Sweep Actions
 - ▶ Sweep Policies

The first three implement maintenance operations of the Advanced Storage Areas. The two “Gen AI” ones maintain IBM Content Assistant Vector Indexing requests, and IBM Content Assistant LLM Operations. They will be discussed in the next subsection.

_14. Click on the **Text Extraction Queue Sweep**.

Queue Sweep: Text Extraction Queue Sweep

General Properties Security Queue Entries

Custom queue sweeps are based on the objects that you want to have processed and the add-ons that you install.

Status: ☒ Enabled ⓘ

• Display name: ⓘ Text Extraction Queue Sweep

Description: ⓘ

Target class: ⓘ Text Extraction Queue Entry

• Sweep action: ⓘ Text Extraction Sweep Action

• Maximum failures: ⓘ 7

Examined object count: ⓘ 5

Processed object count: ⓘ 5

Failed object count: ⓘ 0

Schedule ⓘ

The schedule is the designated periods of the week during which the dispatcher processes sweep requests. If you do not define any periods, the dispatcher runs continuously.

[New](#) [Delete](#)

Start Day	Start Time	Duration
No items to display.		

The “General” tab of this dialog allows to configure a maximum failure count, shows statistic information, and allows to specify a schedule for processing Text Extractions requests. It can be used to configure, for example that text extraction is done only during the night.

_15. Click on the **Properties** tab.

The most important property is the “Maximum Sweep Workers” property. It allows to increase the parallelism with which text extractions are processed. For the “Persistent Text Extraction” it is set to 2.

_16. Click on the **Queue Entries** tab.

The current queue entries can be seen. There should be none. The tab can be used to review failed Text Extraction requests.

_17. Close the **Text Extraction Queue Sweep** window again.

_18. Use the navigation area of ACCE to **navigate** to **Sweep Management -> Sweep Actions** and **click** on **Sweep Actions** to display the defined sweep actions.



	Display Name	Description	Prog ID
☐	Gen AI Queue Sweep Action		com.filenet.engine.genai.sweep.AIQueueSweepHandler
☐	Gen AI Summary Request Job Sweep Action		com.filenet.engine.genai.sweep.SummaryGenerationJobSweepHandler
☐	Gen AI Vector Indexing Request Job Sweep Action		com.filenet.engine.genai.sweep.VectorIndexingJobSweepHandler
☐	Text Extraction Sweep Action	Text Extraction Queue and Job Sweep Handler	com.filenet.engine.extracts.SweepHandler

Text Extraction Sweep Action is a sweep action used in a Custom Sweep Job. Its purpose is to extract text from already uploaded documents of a document class where persistent text extraction is enabled, but no extracted text exists yet. This typically happens when the documents were stored before persistent text extraction was turned on.

The sweep action itself does not perform the text extraction directly. Instead, it identifies the applicable documents and creates an entry for each document in the Text Extraction Queue Sweep. The actual text extraction is then handled by the queue processing.

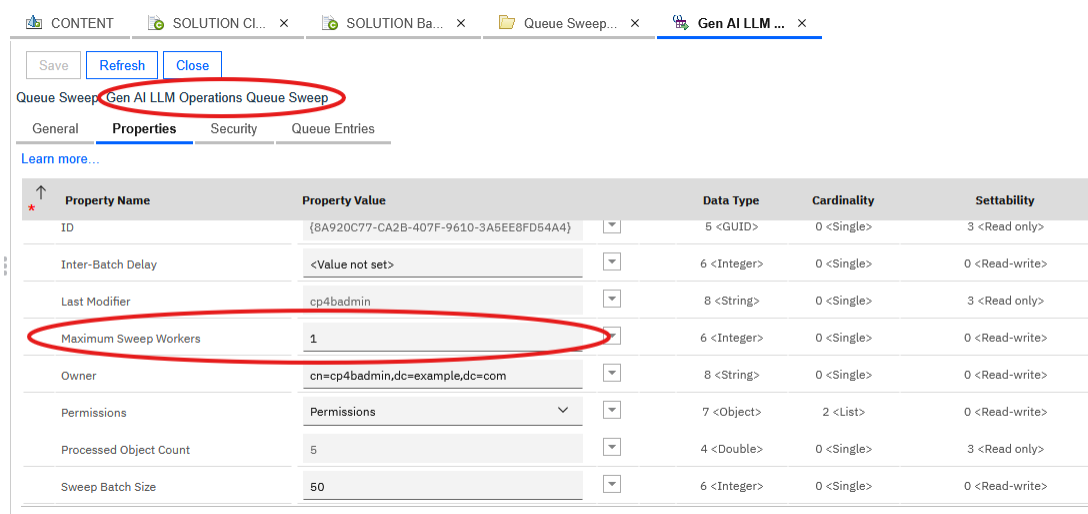
3.2.3 Vector Indexing and Summary Generation

As seen in the last subsection Vector Indexing and Summary Generation for the IBM Content Assistant functionality are performed using a queue sweep.

_1. Use the navigation bar to **navigate** to **Sweep Management -> Queue Sweeps**. Find the following queue sweeps defined:

- **Gen AI LLM Operations Queue Sweep:** This queue sweep handles all queries to the Large Language Model (LLM). This includes but is not limited to requests for summary generation.
- **Gen AI Vector Indexing Queue Sweep:** This queue sweep handles all indexing requests for documents which should be vector indexed, so that they can be added to requests to the LLM.

_2. **Click** on both (one after the other) to bring up a window with the Queue Sweep settings. **Navigate** to the **Properties** tab and **review** the **setting** for the **Maximum Sweep Workers**.

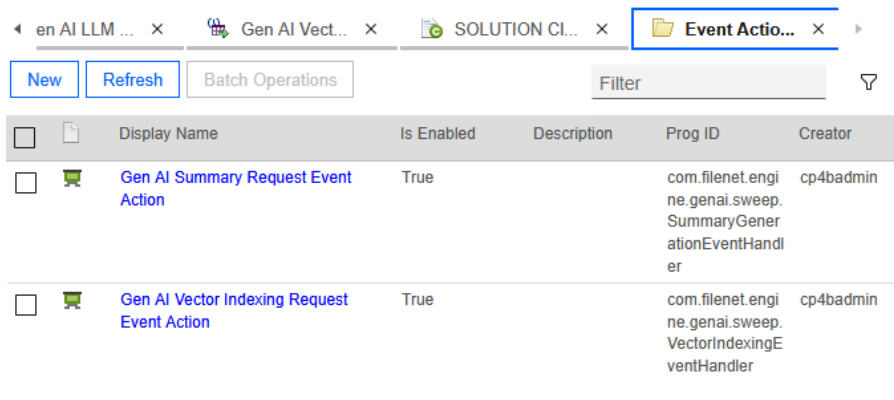


Property Name	Property Value	Data Type	Cardinality	Settability
ID	{8A920C77-CA2B-407F-9610-3A5EE8FD54A4}	5 <GUID>	0 <Single>	3 <Read only>
Inter-Batch Delay	<Value not set>	6 <Integer>	0 <Single>	0 <Read-write>
Last Modifier	cp4badmin	8 <String>	0 <Single>	3 <Read only>
Maximum Sweep Workers	1	6 <Integer>	0 <Single>	0 <Read-write>
Owner	cn=cp4badmin,dc=example,dc=com	8 <String>	0 <Single>	0 <Read-write>
Permissions	Permissions	7 <Object>	2 <List>	0 <Read-write>
Processed Object Count	5	4 <Double>	0 <Single>	3 <Read only>
Sweep Batch Size	50	6 <Integer>	0 <Single>	0 <Read-write>

The “Maximum Sweep Workers” is set to 1, meaning that at most one user question is handled at any given time, to limit the volume of requests to the LLM. The settings can be increased when configuring a separate watsonx.ai entitlement or installing a Customer Managed Software (ICACMS). More details

can be found in the documentation <https://www.ibm.com/docs/en/content-assistant?topic=assistant-adjusting-number-queue-sweep-workers>.

- _3. Use the navigation tree to **navigate** to **Event, Actions, Processes** -> **Event Actions** and **click** on **Event Actions**.



☐	Display Name	Is Enabled	Description	Prog ID	Creator
☐	Gen AI Summary Request Event Action	True		com.filenet.engine.genai.sweep.SummaryGenerationEventHandler	cp4badmin
☐	Gen AI Vector Indexing Request Event Action	True		com.filenet.engine.genai.sweep.VectorIndexingEventHandler	cp4badmin

A document class can be configured for vector indexing and optionally summary generation by subscribing to the corresponding event actions.

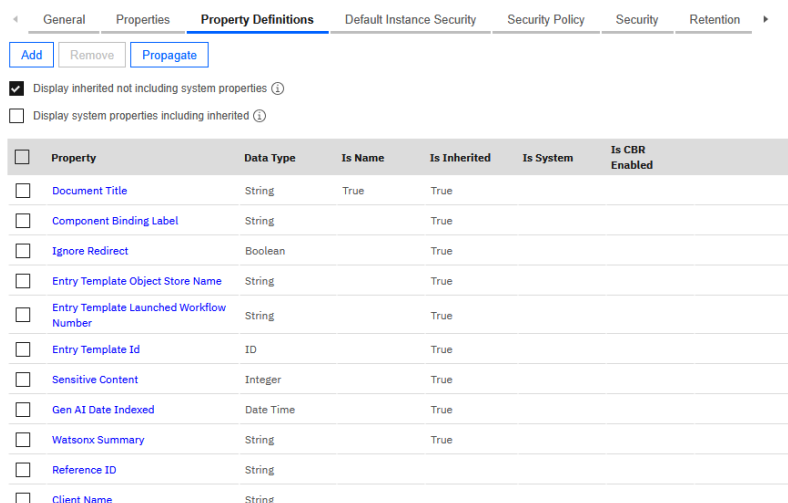
These event actions run synchronously and do not perform the processing themselves. They only create entries in the respective queue sweeps.

In addition, there are sweep actions that enqueue the same requests for existing documents as shown above.

Together, subscriptions (for new documents) and sweep actions (for existing documents) ensure that documents are vector indexed and summaries are generated.

- _4. **Navigate** to **Data Design** -> **Classes** -> **Document** and **click** on the **SOLUTION Client Document** document class. The property window opens.
- _5. **Click** on the **Property Definitions** tab and check the box **Display inherited not including system properties**.

This will show the properties, which are inherited from the “Document” document class.



☐	Property	Data Type	Is Name	Is Inherited	Is System	Is CBR Enabled
☐	Document Title	String	True	True		
☐	Component Binding Label	String		True		
☐	Ignore Redirect	Boolean		True		
☐	Entry Template Object Store Name	String		True		
☐	Entry Template Launched Workflow Number	String		True		
☐	Entry Template Id	ID		True		
☐	Sensitive Content	Integer		True		
☐	Gen AI Date Indexed	Date Time		True		
☐	Watsonx Summary	String		True		
☐	Reference ID	String				
☐	Client Name	String				

Find the two properties “Gen AI Date Indexed” and “Watsonx Summary” there. They are added to the “Document” document class through importing of the “Gen AI Extensions” add-on. The properties are not designed to have their value provided by the user and are write-protected through a Change Preprocessor.

- _6. With the Document class “SOLUTION Client Document” still open, **select** the **Subscriptions** tab.

- _7. **Click** on the **Vector Index subscription** to find out that it “applies to all objects of the class”, executes the “Gen AI Vector Indexing Request Event Action”, subscribes to the “Checkin” and the “Update” events, and is executed “Synchronously”.

The screenshot shows the configuration interface for the 'Gen AI Vector Index' subscription. The 'General' tab is active, showing the display name 'Gen AI Vector Index', description 'Gen AI Vector Index', and target ID 'SOLUTION Client Document'. The 'Event action' is set to 'Gen AI Vector Indexing Request Event Action'. The 'Configuration' tab is also visible, showing options: 'Enabled' (checked), 'Include subclasses' (checked), and 'Synchronous' (checked). The 'Subscribed Events' tab is active, showing a list of available events and selected events. The 'Available Events' list includes 'Classify Complete Event', 'Recover Event', 'Update Security Event', 'Lock Event', and 'Change State Event'. The 'Selected Events' list includes 'Checkin Event' and 'Update Event'.

As explained above, this does not mean that Vector indexing is done during the checkin or update of documents. During the checkin or update of documents, this subscription will merely create an entry in the “Gen AI Vector Indexing Queue Sweep”, which is then scheduled for execution in a separate thread.

- _8. **Navigate** to **Events, Actions, Processes -> Change Preprocessor Actions**. Click on **Change Preprocessor Actions**.

Take notice of the three Gen AI Change Preprocessors, which were imported through the IBM Content Assistant add-on. Among other things, they protect IBM Content Assistant metadata in the documents and the Object Store from unauthorized changes.

The screenshot shows the 'Change Preprocessor Actions' table in the IBM Content Assistant interface. The table has columns for 'Display Name' and 'Is Enabled'. There are three rows of data, each with a checkbox in the first column and a link in the second column.

	Display Name	Is Enabled
☐	Gen AI Configuration Change Preprocessor Action	True
☐	Gen AI Document Change Preprocessor Action	True
☐	Gen AI Query Change Preprocessor Action	True

4 Exercise Application Programming Interface (API)

4.1 Introduction

Both statements are correct:

- IBM Content Assistant does not itself expose an Application Programming Interface (API).
- IBM Content Assistant can be used in custom applications to enrich them with AI, optionally adding documents to provide context to queries.

How can that be?

IBM Content Assistant does not provide an Application Programming Interface itself. But it can be used through the query classes, which are defined by the IBM Content Assistant add-on in an Object Stores for which it was enabled. Within this exercise, those classes will be shown and used to run queries.

Consequently, the first part of the exercise will be performed in ACCE, which is one place where query objects can be created. Later parts of the exercise demonstrate that the same kind of query objects can be created through GraphQL as well.

The third option would be to use the CPE Java APIs to create the query objects. Yet, it is out of scope for this lab to create and run Java programs. Refer to the [documentation](#) for additional information.

4.2 Exercise Instructions

4.2.1 Base Query Class

The classes, which are designed to allow raising queries in an IBM Content Assistant enabled Object Store are described in the documentation on page “GenAI query classes” <https://www.ibm.com/docs/en/content-assistant?topic=applications-genai-query-classes>. The documentation lists the metadata for each of the classes.

- _1. **Login to ACCE** with the user account which was provided to you.
- _2. **Open the CONTENT** Object Store.
- _3. **Navigate to Data Design -> Classes -> Other Classes -> Abstract Persistable** and **click** on the **Gen AI Base Query** class.

The properties of this class can be found on the Property Definitions tab, see <https://www.ibm.com/docs/en/content-assistant?topic=classes-genaibasequery> for details.

The screenshot shows the IBM Administrative Console for Content Platform Engine. The left sidebar displays a tree view of the 'CONTENT' Object Store, with 'Gen AI Base Query' selected under 'Other Classes'. The main panel shows the 'Class Definition: Gen AI Base Query' with the 'Property Definitions' tab active. It lists several properties with their data types and inheritance status.

Property	Data Type	Is Name	Is Inherited	Is S
Gen AI LLM Prompt	String			
Gen AI LLM Model Name	String			
Gen AI Max Output Tokens	Integer			
Gen AI Prompt Template	String			
Gen AI LLM Response	String			

On the left side, the class hierarchy below the base query class can be unfolded. The class names “Gen AI Adhoc Summary” and “Gen AI Document Comparison” are more or less self-explanatory. The “Gen AI Vector Query” class is the base class for queries that are executed against one or multiple documents attached to them from the Vector database, hence its name.

- _4. From the **Actions** menu on this class, **invoke** the **Create instance** function. The Object Properties window opens showing the (still empty) metadata fields for the new query object.

Object Properties

Specify the property values for the object.

↓ *	Property Name	Property Value	Data Type
	Replication Group	<Value not set>	7 <Object>
	Gen AI LLM Prompt	<Value not set>	8 <String>
	Gen AI LLM Model Name	<Value not set>	8 <String>
	Gen AI Max Output Tokens	<Value not set>	6 <Integer>
	Gen AI Prompt Template	<Value not set>	8 <String>
	Gen AI LLM Response	<Value not set>	8 <String>

- _5. To run a query (in this case against the IBM watsonx.ai llama-4-maverick LLM), **fill out** the property **Gen AI LLM Prompt**, for example with the query **What is ECM?**

With the default configuration of IBM Content Assistant, which is using a watsonx.ai account, a selection of LLMs is available. One is the IBM watsonx Granite foundation model. See the section “Changing the Content Assistant large language model for an object store”

<https://www.ibm.com/docs/en/content-assistant?topic=ccaspos-changing-content-assistant-large-language-model-object-store> for details.

If a separate account is configured, and that account has access to other LLMs as well, the LLM to be chosen can be configured in the “Gen AI LLM Model Name” property.

< Back Next > Finish Cancel

Object Properties

Specify the property values for the object.

↓ *	Property Name	Property Value	
	Replication Group	<Value not set>	▼
	Gen AI LLM Prompt	What is ECM?	▼
	Gen AI LLM Model Name	<Value not set>	▼
	Gen AI Max Output Tokens	<Value not set>	▼
	Gen AI Prompt Template	<Value not set>	▼
	Gen AI LLM Response	<Value not set>	▼

- _6. **Click** on **Next**, then **Finish**. Notice that the request takes quite some time to complete. Finally, the “Success” page is shown, here after 24 seconds.

Open Close

Success

Start time: September 9, 2025 at 9:27:44 AM Central European Summer Time

End time: September 9, 2025 at 9:28:07 AM Central European Summer Time

Elapsed time: 24 seconds

100%

Created "GenaiBaseQuery"

The reason (probably) is that ACCE might save the object with the “Refresh” flag enabled. This will try to read the updated values for the fields after creating the object, which includes trying to read the “Gen AI LLM Response” field. This will implicitly wait for the new entry “Gen AI LLM Operations Queue Sweep” to be processed.

In a custom application, if the call for creating a query is to be return immediately, then it would be recommended to unset the “Refresh” flag, see <https://www.ibm.com/docs/en/content-assistant?topic=applications-handling-genai-query-objects> for more details.

_7. **Click on Open**, which opens the new object with the “Properties” tab. **Find** the updated field **Gen AI LLM Response** with the answer to the query. Click on the icon next to the field and

Gen AI Max Output Tokens	<Value not set>	▼	Display or Edit Value
Gen AI LLM Model Name	<Value not set>	▼	
Gen AI LLM Prompt	What is ECM?	▼	
Gen AI LLM Response	ECM DefinitionThe term ECM can refer to different	▼	ECM DefinitionThe term ECM can refer to different concepts depending on the context. Enterprise Content Management (ECM): A set of technologies used to capture, manage, store, preserve, and deliver content and documents related to organizational processes. Extracellular Matrix (ECM): A three-dimensional network of extracellular macromolecules and minerals, such as collagen, enzymes, and glycoproteins, that provide structural and biochemical support to surrounding cells. Without more context, it's challenging to determine which definition is most relevant.
Gen AI Prompt Template	<Value not set>	▼	

Do you recall why the response is using HTML tags? Find the answer in section 3.2.1 Content Assistant Add-on and Configuration in step _14.

_8. **Click on Search** in the navigation. The “Saved Searches” window opens on the right side.

_9. **Click on New Object Store Search.**

_10. In the “New Object Store Search” window, **select** the **Gen AI Base Query** class, and **click** on **Run** to run the search. **Dismiss** the **warning dialog**.

A list of prior query objects to which the logged-on user has access appears in the Search Results window as shown in the example below:

Search: New Object Store Search		
Simple View	SQL View	Bulk Actions (Disabled)
Search Results X		
Actions ▼		
☐ ID	Class Description	Is Marked For Deletion
☐ {A0E3D097-0000-C510-A6AA-99793DC8F3D1}	Gen AI MultiDocument Q	False
☐ {405F3698-0000-C51E-96E8-D398EA3E03DB}	Gen AI MultiDocument Q	False
☐ {D0583D98-0000-C311-BD0F-05E85887AE32}	Gen AI MultiDocument Q	False
☐ {6019AC98-0000-CD1D-AA2D-BEC4CE55CD72}	Gen AI Base Query	False

By default, FileNet retains a history of the searches, which were performed by the logged-on users. This history can be seen in the IBM Content Assistant chat window in IBM Content Navigator, too. There are a couple possibilities to fine-tune this behavior according to the needs of customers:

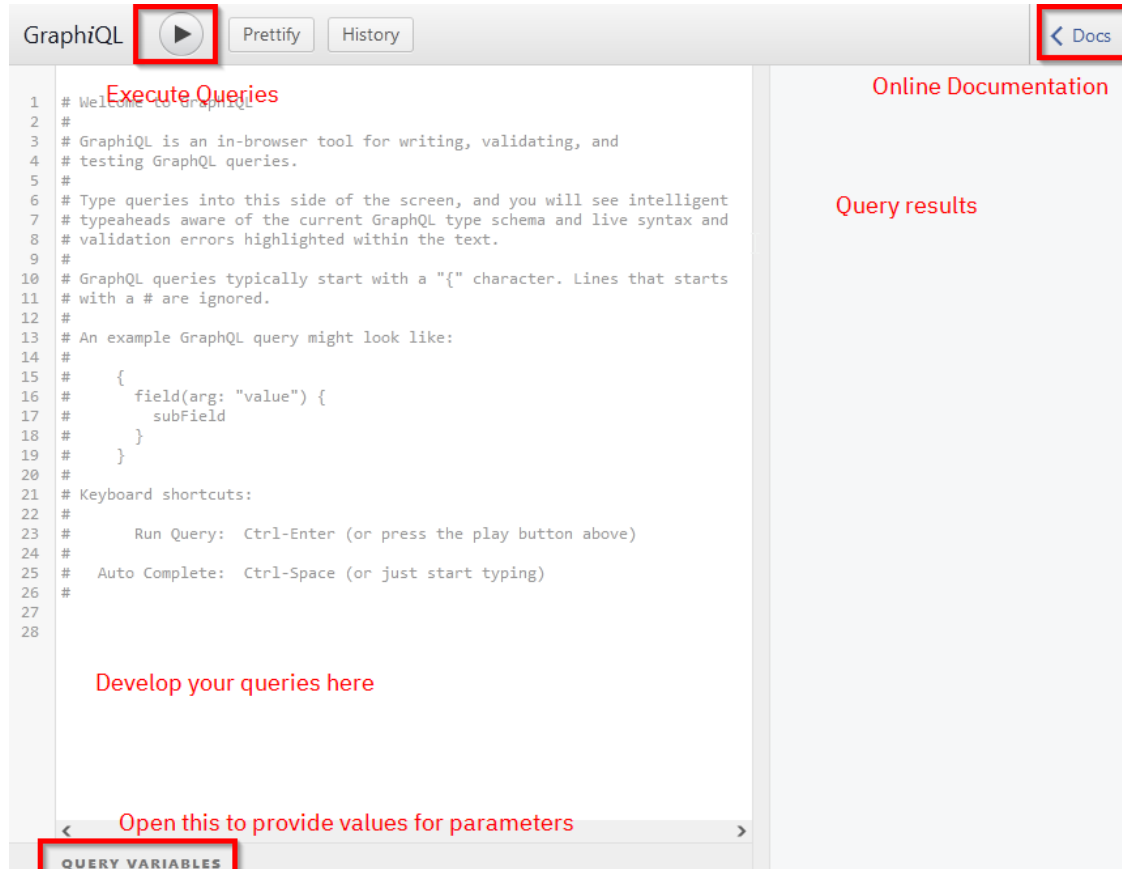
- The “Default Instance Permission” of the “Gen AI Base Query” class can be modified to only include permissions for the “#CREATOR-OWNER”. Then the history of queries would only be visible to the user who raised them.
- A Disposal Policy can be defined for “Gen AI Base Query” objects, to automatically delete such objects after a defined time span. See “Disposal Policies” in the FileNet Content Manager documentation for details <https://www.ibm.com/docs/en/filenet-p8-platform/5.7.0?topic=development-disposal-policies>.
- The history could be summarized through a Background search to derive information for invoicing users based on the Content Assistant queries they have raised.

4.2.2 Using Base Query Class from GraphQL

Instead of using ACCE, the queries from this and further subsections can also be executed from the Content Services GraphQL UI. This approach allows for easier embedding of ICA capabilities into custom applications.

_1. **Login** to the **Content Services GraphQL UI** with the URL and user account which was provided to you.

For a compact introduction, write the queries or mutations into the pane in the upper left corner. The existing sample can be removed before. The field in the lower left corner contains values for variables appearing in the query. By clicking the round play button (Execute Query Button), the query can be executed. Query results are displayed on the right side.



For running a query, new objects for running LLM queries are created in FileNet Content Engine using one of the IBM Content Assistant Query classes.

The first example will run the same query as the one from the last subsection.

_2. **Remove** any **query** shown when opening the GraphQL window.

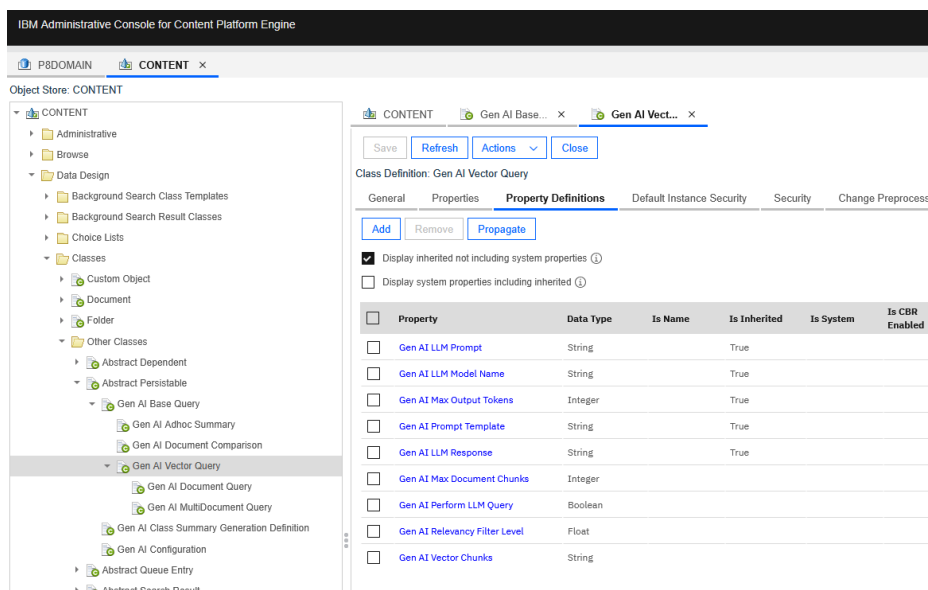
_3. **Copy & Paste** below **query** into the field in the upper left corner and **click** on the **Execute Query Button**.

```
mutation basequery {
  whatisecm: createCmAbstractPersistable(
    repositoryIdentifier: "CONTENT",
    classIdentifier: "GenAIBaseQuery",
    cmAbstractPersistableProperties: {
      properties:
        [ { GenaiLLMPrompt: "What is ECM?" } ]
    }
  )
  {
    properties(includes: ["GenaiLLMPrompt", "GenaiLLMResponse" ])
      { alias value }
  }
}
```

See Content Lab 2 “Interfacing FileNet Content Platform Engine with GraphQL on Cloud Pak for Business Automation” for an introduction into FileNet Content Services GraphQL.

In this mutation, the request “createCmAbstractPersistable” can be used to create any subclass of the base class “Abstract Persistable” in FileNet Content Engine. The mutation request is provided with the Object Store, the class name, and the properties of the new object. In this case this is only the GenaiLLMPrompt property. The lower part describes which components from the newly created object shall be contained in the result. In this case its only two properties, and from them only name and value. Finally, the label “whatisecm” is configured to be used as the name of the response in the returned JSON information. This way, several such mutations can be sent to GraphQL within one request, if needed, for improved efficiency over wide area network connections.

- _4. Queries against the complete vector database can be made in much the same way. Look up the class “Gen AI Vector Query” in ACCE on the Property Definitions tab to get an overview of its properties. Enable “Display inherited not including system properties” to have the display include the inherited properties from the base class.



To use the class, the property “Gen AI LLM Prompt” needs to be provided with the prompt for the LLM to answer, as before. Additionally, the property “Gen AI Perform LLM Query” needs to be set to true. The query can be configured further by passing other configuration properties, see the description of this query class in the IBM Content Assistant documentation <https://www.ibm.com/docs/en/content-assistant?topic=classes-genaivectorquery> for details.

However, on the execution of such queries, only documents available to the logged-on user will be used.

- _5. Use below sample to perform such a query. Compare the query with the previous query using the GenaiBaseQuery class. There are only few differences, which are typeset here in bold.

```
mutation vectorquery {
  bautomationelite: createCmAbstractPersistable(
    repositoryIdentifier: "CONTENT",
    classIdentifier: "GenaiVectorQuery",
    cmAbstractPersistableProperties: {
      properties:
        [ { GenaiLLMPrompt: "Summarize the information available for Automation Elite  
Inc. Since when are we making business with them?" },
          { GenaiPerformLLMQuery: true } ]
    }
  )
  {
    properties(includes: ["GenaiLLMPrompt", "GenaiLLMResponse" ])
      { alias value }
    }
  }
}
```


4.2.3 Queries with One or More Documents

For most other kind of queries, one or more documents need to be passed as references, to be stored in properties of type OBJECT in the IBM Content Assistant query classes. As a blueprint for this kind of queries, the first example creates an LLM query with one document passed to it.

1. For running this kind of query, information to the referenced document needs to be provided, i.e. the Object ID of the document, and the Object ID or the name of the Document class. The following GraphQL query lists the documents, which are stored in the Folder “SOLUTION Client Onboarding”. **Copy & Paste the query to GraphQL.**

```
query solutionDocuments {
  solutiondocuments: folder(
    repositoryIdentifier: "CONTENT",
    identifier: "/SOLUTION Client Onboarding") {
    name
    containedDocuments {
      documents {
        id name className
      }
    }
  }
}
```

2. **Copy the result into an editor window.** On a Windows System you might invoke Notepad and copy the result into the Notepad window on a new tab. Use similar tools on computers with different operation systems.

Another solution would be to leave the browser tab with the output from GraphQL open in the browser for future reference and use a new tab for the following queries.

3. Before using the query, review the class definition of the GenaiDocumentQuery again in ACCE. Go to the Property definitions tab and include display of inherited properties, to get an overview over which parameters can / must be provided. As the class inherits from the GenaiVectorQuery document class, find its properties among the inherited ones. The only new one which needs to be provided is the GenaiContextDocument property.

The screenshot shows the IBM Administrative Console for Content Platform Engine. The left sidebar displays a tree view of the object store 'CONTENT', with 'Gen AI Document Query' selected under 'Other Classes'. The main panel shows the 'Class Definition: Gen AI Document Query' with tabs for 'General', 'Properties', 'Property Definitions', 'Default Instance Security', 'Security', and 'Change'. The 'Property Definitions' tab is active, showing a table of properties. The 'Display inherited not including system properties' checkbox is checked. The table lists various properties with their data types and inheritance status.

Property	Data Type	Is Name	Is Inherited	Is System
Gen AI LLM Prompt	String		True	
Gen AI LLM Model Name	String		True	
Gen AI Max Output Tokens	Integer		True	
Gen AI Prompt Template	String		True	
Gen AI LLM Response	String		True	
Gen AI Max Document Chunks	Integer		True	
Gen AI Perform LLM Query	Boolean		True	
Gen AI Relevancy Filter Level	Float		True	
Gen AI Vector Chunks	String		True	
Gen AI Context Document	Object			

- _4. **Copy & Paste** the below the **parameterized GraphQL query** into the **query pane** of the GraphQL browser window for running a single document query. For providing the value of the GenaiLLMPrompt property, a String parameter is used, for the GenAIContextDocument property, as an Object is expected (see above), the parameter type is ObjectReferenceInput in Graph QL.

Similar as for the Vector Query, for the Document query the property “GenaiPerformLLMQuery” needs to be set to true. By passing “false” here, developers of custom applications can test their development of an IBM Content Assistant integration in custom applications, without actually executing the queries.

```
mutation documentquery($query: String!, $doc: ObjectReferenceInput!) {
  singleDocQuery: createCmAbstractPersistable(
    repositoryIdentifier: "CONTENT",
    classIdentifier: "GenaiDocumentQuery",
    cmAbstractPersistableProperties: {
      properties: [
        { GenaiLLMPrompt: $query },
        { GenaiPerformLLMQuery: true },
        { GenaiContextDocument: $doc }
      ]
    }
  )
  {
    properties(includes: ["GenaiLLMPrompt", "GenaiLLMResponse"])
    { alias value }
  }
}
```

- _5. For running the query, in the lower left corner of the GraphQL UI, the parameter values need to be passed using JSON syntax. In the example below **replace DOCID** first with the **ID of the Banking Information Document for Automation Elite Inc.** then with the **ID of the Banking Information Document for Legacy Consulting.**

```
{
  "query": "Is the document a Banking Information statement for the company Automation Elite Inc? Answer with True or False only.",
  "doc": {
    "repositoryIdentifier": "CONTENT",
    "classIdentifier": "SOLUTION_Bank_Information",
    "identifier": "DOCID"
  }
}
```

- _6. As the next example in this section, a GenaiMultiDocumentQuery will be executed. Look up the class definition in ACCE first, as before. It looks very similar to the GenaiDocumentQuery class.

The screenshot shows the IBM Administrative Console for Content Platform Engine. On the left, a tree view displays the class hierarchy under 'CONTENT'. The 'Gen AI MultiDocument Query' class is selected. The main panel shows the 'Property Definitions' tab for this class. It includes a table of properties with columns: Property, Data Type, Is Name, Is Inherited, and Is System.

Property	Data Type	Is Name	Is Inherited	Is System
Gen AI LLM Prompt	String		True	
Gen AI LLM Model Name	String		True	
Gen AI Max Output Tokens	Integer		True	
Gen AI Prompt Template	String		True	
Gen AI LLM Response	String		True	
Gen AI Max Document Chunks	Integer		True	
Gen AI Perform LLM Query	Boolean		True	
Gen AI Relevancy Filter Level	Float		True	
Gen AI Vector Chunks	String		True	
Gen AI Context Documents	ID			

- _7. **Click** on the **Gen AI Context Documents** property. A window with details about the “Property Definition” opens. **Verify** that the **Cardinality** of the property is given as **Multi** which means that an Array of Document IDs need to be passed for the property value. **Close** the Property Definition **window**.

Note that a description of the property is also available there. Note further that the symbolic name of that property is “GenaiContextDocuments”. Within applications, always the symbolic names are used. The display names are not only subject to change, but they can also appear translated, when switching the browser to a different language.

Property Definition

General	Alias IDs	More	Modification Access
Display name: ⓘ	Gen AI Context Documents		
Symbolic name: ⓘ	GenaiContextDocuments		
Description: ⓘ	The documents to use as context for the query		
Data type: ⓘ	ID ▾		
Cardinality:	Multi ▾		
Primary Id:	{A9A962D6-2589-446F-854C-17DDE45280FF}		
Used in classes: ⓘ	▾		
☐ Is system owned			

- _8. Below GraphQL mutation will be creating a GenaiMultiDocumentQuery object. It is again parameterized, as the GenaiDocumentQuery. The type of the second parameter was adapted to provide an array of Document IDs. **Copy & Paste** the **query** into the GraphQL query window.

```
mutation multidocumentquery($query: String!, $docs: [ ID ]) {
  multiDocQuery: createCmAbstractPersistable(
    repositoryIdentifier: "CONTENT",
    classIdentifier: "GenaiMultiDocumentQuery",
    cmAbstractPersistableProperties: {
      properties:
        [ { GenaiLLMPrompt: $query },
          { GenaiPerformLLMQuery: true },
          { GenaiContextDocuments: $docs }
        ]
    }
  )
  {
    properties(includes: ["GenaiLLMPrompt", "GenaiLLMResponse"])
    { alias value }
  }
}
```

- _9. The parameter values are again provided using JSON. **Replace** the **two Object IDs** from the Banking Information Documents of the two companies.

```
{
  "query": "What companies do we have banking information for? Provide bank names and the date of the banking information.",
  "docs": [
    "DOCID_AUTOMATION_ELITE",
    "DOCID_LEGACY_CONSULTING"
  ]
}
```

_10. For the comparison of two documents, the Gen AI add-on defines the “Gen AI Document Comparison” subclass of the Gen AI Base query. It has following properties

Class Definition: Gen AI Document Comparison

General Properties **Property Definitions** Default Instance Security Security Change Preprocessor

Add Remove Propagate

☒ Display inherited not including system properties ⓘ
☐ Display system properties including inherited ⓘ

☐	Property	Data Type	Is Name	Is Inherited	Is System	Is CBR Enabled
☐	Gen AI LLM Prompt	String		True		
☐	Gen AI LLM Model Name	String		True		
☐	Gen AI Max Output Tokens	Integer		True		
☐	Gen AI Prompt Template	String		True		
☐	Gen AI LLM Response	String		True		
☐	Gen AI Context Document	Object				
☐	Gen AI Context Document 2	Object				
☐	Gen AI Additional Instructions	String				

_11. Below GraphQL mutation will be creating a GenAIDocumentComparison object. It is again parameterized, as the GenaiDocumentQuery. The type of the parameter is the same as for the GenaiDocumentQuery, just two of them are needed for the two documents. And there is an additional field for the additional instructions. **Copy & Paste** the **query** into the GraphQL query window.

```
mutation multidocumentquery($instructions: String!,
                             $doc1: ObjectReferenceInput!,
                             $doc2: ObjectReferenceInput!) {
  compare: createCmAbstractPersistable(
    repositoryIdentifier: "CONTENT",
    classIdentifier: "GenaiDocumentComparison",
    cmAbstractPersistableProperties: {
      properties:
        [ { GenaiLLMPrompt: "Compare the documents" },
          { GenaiContextDocument: $doc1 },
          { GenaiContextDocument2: $doc2 },
          { GenaiAdditionalInstructions: $instructions }
        ]
    )
  {
    properties(includes: ["GenaiLLMPrompt", "GenaiLLMResponse",
                          "GenaiAdditionalInstructions"])
    { alias value }
  }
}
```

_12. The parameter values require substitutions of the Document IDs again. Provide the two banking information documents.

```
{
  "instructions": "List only the similarities",
  "doc1": {
    "repositoryIdentifier": "CONTENT",
    "classIdentifier": "SOLUTION_Bank_Information",
    "identifier": "DOCID_AUTOMATION_ELITE"
  },
  "doc2": {
    "repositoryIdentifier": "CONTENT",
    "classIdentifier": "SOLUTION_Bank_Information",
    "identifier": "DOCID_LEGACY_CONSULTING"
  }
}
```

_13. Finally, try to create a parameterized GraphQL query for invoking the GenaiAdhocSummary query class, by looking up the class definition in ACCE, and adapting the example for the GenaiMultiDocumentQuery. For that class only the documents need to be supplied, in the same way as for the GenaiMultiDocumentQuery, the values for the GenaiPerformLLMQuery and the GenaiLLMPrompt can be removed.

Find the solution at the end of this document, following the Troubleshooting section.

Congratulations you have successfully completed the lab
“Configuration and Usage of IBM Content Assistant on Cloud Pak for Business Automation”!

5 Troubleshooting

5.1 GENAI_QUERY_FAILED

When the connection between the FileNet environment and the IBM Content Assistant Server in the SaaS environment is unavailable or overloaded, the following error might be observed in the GraphQL GUI:

```
"errorCode": "FNRJG1005",
"message": "Error when calling the Content Platform Engine API.",
"explanation": "An error occurred when calling the Content Platform Engine API. An exception occurred in the handler. Message was: GENAI_QUERY_FAILED: Vector query failed. The response is: {\"message\": \"Endpoint request timed out\"}\",
"userResponse": "Contact your system administrator for assistance.",
"statusCode": "500",
"serverErrorMessage": "An exception occurred in the handler. Message was: GENAI_QUERY_FAILED: Vector query failed. The response is: {\"message\": \"Endpoint request timed out\"}\",
"classification": "DataFetchingException"
```

Retrying the request has proved to be a good strategy to handle this error situation.

5.2 Body has already been consumed.

```
Response.text: Body has already been consumed.
```

This error can happen when the authentication token of the CP4BA environment has expired. To resolve the situation, place the mouse cursor on the address bar of the browser, and hit Return, to reload the page. Pressing F5 key on a Windows machine will have the same result.

If the login page of Cloud Pak for Business Automation appears as a result, then the problem was the Authentication token. Use the provided username and password to login again. Your query and parameter values should still be there after logging in again.

5.3 Subscription contract limits

```
Subscription contract limit for the number of users has exceeded, additional packages can be purchased.
```

If you get this message, the problem is a limitation configured on the IBM Content Assistant server. Many kinds of limitations are imposed there. Configured usage limits can be increased by the IBM Content Assistant administrator.

If you get this error, reach out to the instructors of the event for assistance.

6 Sample GraphQL for the GenaiAdhocSummary query class

Below GraphQL query implements the requirement. Compared to the GenaiMultiDocumentQuery, the parameter for the query was removed, as well as the two property definitions, which are not needed.

```
mutation multidocumentquery($docs: [ ID ]) {  
  adhocsummary: createCmAbstractPersistable(  
    repositoryIdentifier: "CONTENT",  
    classIdentifier: "GenaiAdhocSummary",  
    cmAbstractPersistableProperties: {  
      properties:  
        [{ GenaiContextDocuments: $docs }]  
    }  
  )  
  {  
    properties(includes: ["GenaiLLMPrompt", "GenaiLLMResponse"])  
    { alias value }  
  }  
}
```

For the parameters, only remove the prompt, for example:

```
{  
  "docs": [  
    "DOCID_AUTOMATION_ELITE",  
    "DOCID_LEGACY_CONSULTING"  
  ]  
}
```